

Chapter 94

Color Nodes

This chapter details the Color nodes available in Fusion. The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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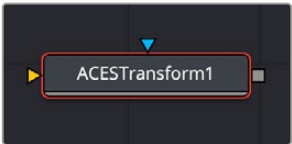
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ColorFX

A sub category of the Color tools in Fusion that gives you access to various color processing tools.

ACESTransform [ATr]



The ACES Transform node

ACES Transform Node Introduction

A simple node that lets you perform color transforms using the ACES Input Device Transform and ACES Output Device Transform parameters.

Inputs

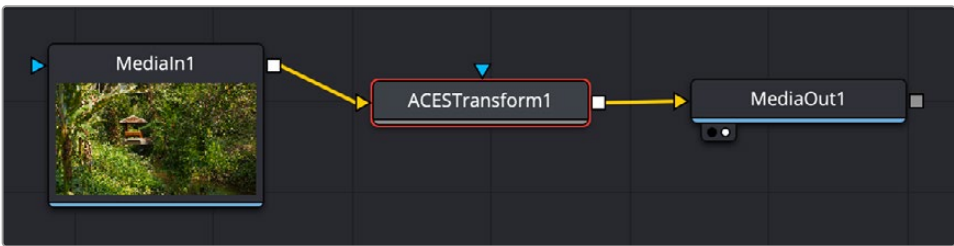
The two inputs on the ACES Transform node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

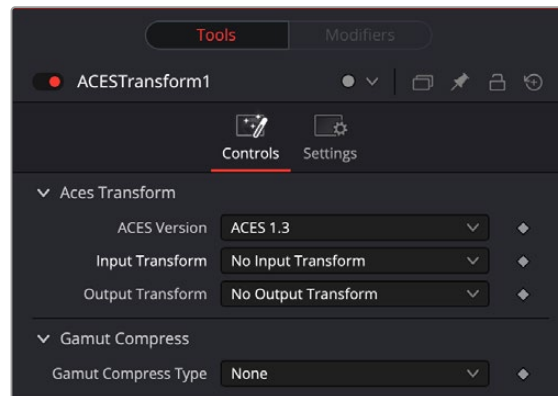
Basic Node Setup

The ACES Transform node, like many 2D image-processing nodes, receives a 2D image, such as a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



An ACES Transform node applied to a MediaIn1 node

Inspector



ACES Transform controls

Controls Tab

The Controls tab contains the few primary controls necessary for performing the ACES transform on the input.

ACES Version

Lets you choose which version of ACES you want to use.

Input Transform

This menu lets you choose which IDT (Input Device Transform) to use to process the image. Typically you would pick the camera that your footage was shot with here.

Output Transform

This menu lets you choose an ODT (Output Device Transform) with which to transform the image data to the color space for your exported timeline.

Gamut Compress Type

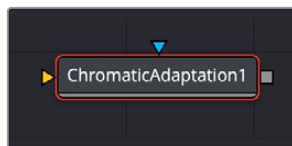
This lets you choose the type of ACES Gamut compression applied to the node. Choosing either option can prevent monochromatic highly saturated light sources (LEDs, neon signs, tail lights, etc.) from clipping the gamut.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Chromatic Adaptation [CRa]



The Chromatic Adaptation node

Chromatic Adaptation Node Introduction

A simple node that lets you precisely transform an image that has been lit or processed assuming one specific color temperature to another user-selectable color temperature. This transformation alters the appearance of all colors in the image as perceived by the human vision system in the same way that a new illuminant would, whether that illuminant is a light in the environment being recorded or the color temperature of a display on which the image is shown. This node is useful for performing specific color temperature transformations as part of color management workflows or for setting up precise color temperature adjustments.

Inputs

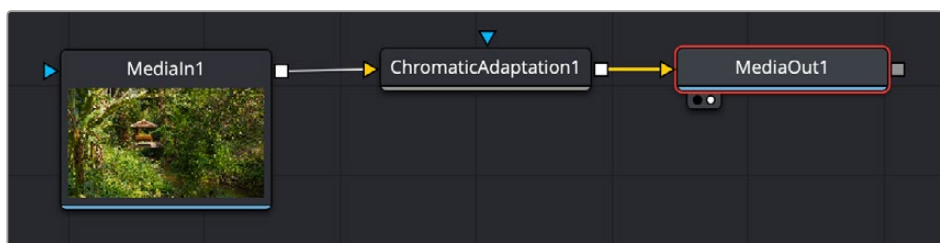
The two inputs on the Chromatic Adaptation node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

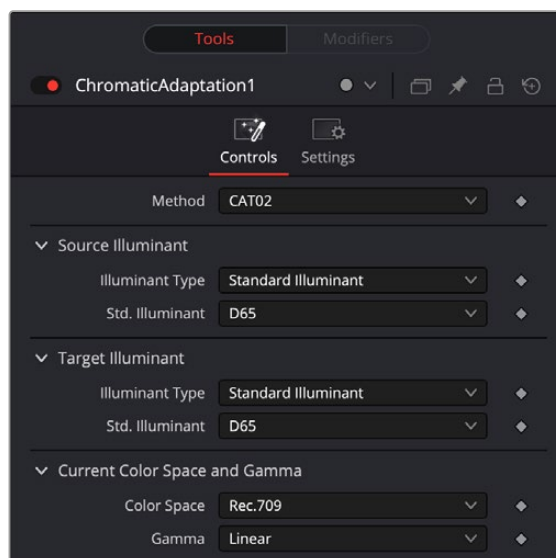
Basic Node Setup

The Chromatic Adaptation node, like many 2D image-processing nodes, receives a 2D image, such as a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Chromatic Adaptation node applied to a MediaIn1 node

Inspector



ACES Transform controls

Controls Tab

The Controls tab contains the primary controls necessary for adjusting the Chromatic Adaptation parameters.

Method

The Method pop-up menu provides a variety of different transform methods to choose from, defaulting to CAT02. Each option in the Method pop-up menu uses different measurement datasets to create individual CAT matrixes to guide this transformation. As a result, each method prioritizes different levels of accuracy for different sets of colors. For example:

- CAT02 has a non-linear component that compensates for the tendency of extremely saturated blues to go purple, a typical weakness of other methods. It usually gives the best result for the widest variety of measured data sets and works best for emissive sources (displays) and dim viewing environments.
- Bradford Linear is also a commonly used method, albeit one in which extremely saturated blues will go purple during the transform. It works well for both emissive sources in dim environments and for reflective sources (theater screens) and dark environments.
- Von Kries is one of the oldest methods in common use, although it's also one in which extremely saturated blues will go purple during the transform. This, as well as all other methods, are available if you need to match work done in another image processing application.

NOTE: Be aware that all methods listed will match neutral colors perfectly; the only differences lie in how different ranges of saturated color are transformed.

Source/Target Illuminant

You control this transformation by using pop-up menus to define the Illuminant Type of the source (typically the color temperature the camera was set to) and that of the target that you want to transform the image to. For both sets of controls, you can choose a Standard Illuminant from a list, a Color Temperature via a slider, or CIE 1931 xy coordinate values. This image transform is extremely precise because the image is first transformed from the Timeline Color Space to XYZ, and then it's transformed to match the LMS (long, medium, short) color space that models the cone response of the human eye to colors lit by different illuminants.

Current Color Space and Gamma

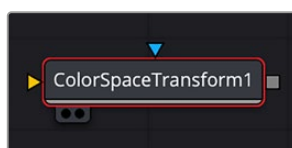
This node also takes into account the current color space and gamma of the clip, which default to those set for the current timeline. If you wish to change these values, you can do so using the Color Space and Gamma menus.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Color Space Transform [CSt]



The Color Space Transform node

Color Space Transform Node Introduction

A simple node that lets you perform the kind of color transforms that LUTs do, but instead of using lookup tables, this plugin uses the same math used by Resolve Color Management (RCM) in order to do extremely clean color transforms without clipping.

Inputs

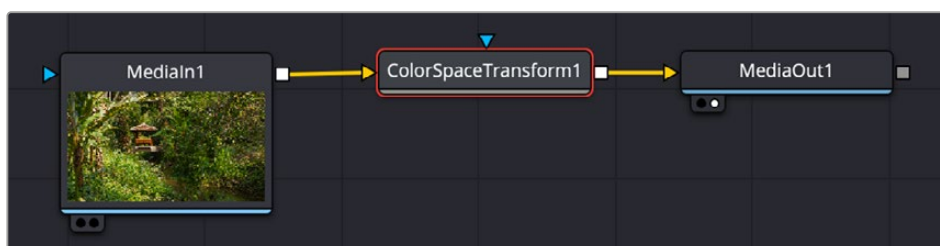
The two inputs on the Color Space Transform node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

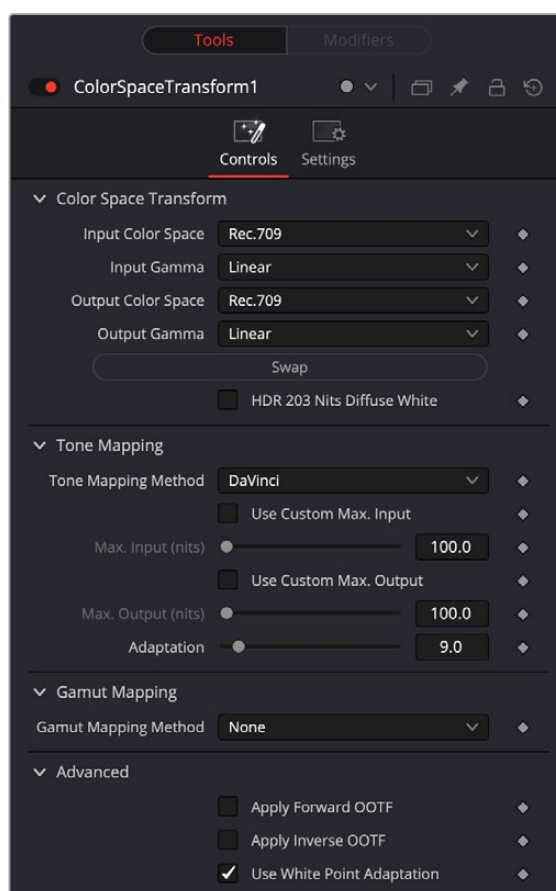
Basic Node Setup

The Color Space Transform node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Color Space Transform node applied to a MediaIn1 node

Inspector



Color Space Transform controls

Controls Tab

The Controls tab contains the primary controls necessary for performing the Color Space Transform on the input.

Color Space Transform

Exposes four pop-up menus that let you set an Input Colorspace, Input Gamma, Output Colorspace, and Output Gamma, in order to do controlled transforms from the Input settings to the Output settings, right within a node. The Swap button lets you quickly reverse the color and gamma space conversion.

Tone Mapping

Tone Mapping lets you enable tone mapping to accommodate workflows where you need to transform one color space into another with a dramatically larger or smaller dynamic range by automating an expansion or contraction of image contrast in such a way as to give a pleasing result with no clipping.

- **None:** This setting disables Input DRT Tone Mapping. No tone mapping is applied to the Input to Timeline Color Space conversion at all, resulting in a simple 1:1 mapping to the Timeline Color Space.
- **Clip:** Hard clips all out-of-bounds values.
- **Simple:** Uses a simple curve to perform this transformation, compressing or expanding the highlights and/or shadows of the timeline dynamic range to better fit the output dynamic range. Note that the “Simple” option maps between approximately 5500 nits and 100 nits, so if you’re mapping from an HDR source with more than 5500 nits to an SDR destination there may still be some clipping of the highlights above 5500 nits.

- **Luminance Mapping:** Same as DaVinci, but more accurate when the Input Color Space of all your media is in a single standards-based color space, such as Rec. 709 or Rec. 2020.
- **DaVinci:** This option tone maps the transform with a smooth luminance roll-off in the shadows and highlights, and controlled desaturation of image values in the very brightest and darkest parts of the image. This setting is particularly useful for wide-gamut camera media and is a good setting to use when mixing media from different cameras.
- **Saturation Preserving:** This option has a smooth luminance roll-off in the shadows and highlights but does so without desaturating dark shadows and bright highlights, so this is an effective option for colorists who like to push color harder. However, because over-saturation in the highlights of the image can look unnatural, two parameters are exposed to provide some user-adjustable automated desaturation.

Sat. Rolloff Start: Lets you set a threshold, in nits (cd/m²), at which saturation will roll off along with highlight luminance. Beginning of the rolloff.

Sat. Rolloff Limit: Lets you set a threshold, in nits (cd/m²), at which the image will be totally desaturated. End of the rolloff.

Use Custom Max Input/Output: Checking these boxes and adjusting the slider below allows you to specify the minimum and maximum luminance of the input image in nits. Using these two sliders together, you can set which value from the Input Gamma is mapped to which value of the Output Gamma.

- **Adaptation:** Used to compensate for large differences in the viewer's state of visual adaptation when viewing a bright image on an HDR display versus seeing that same image on an SDR display. For most "average" images this setting works best set between 0–10. However, when you're converting very bright images (for example, a snow scene at noon), then using a higher value will yield more image detail within the highlights.

Gamut Mapping

Gamut Mapping Method accommodates workflows where you need to transform one color space into another with a dramatically larger or smaller gamut by helping to automate an expansion or contraction of image saturation in such a way as to give a pleasing and naturalistic result with no clipping.

- **None:** results in no Gamut mapping at all.
- **Saturation Mapping:** Enables saturation mapping to fit the range of saturation values from the Input Color Space and Gamma into the Output Color Space and Gamma. It enables the Saturation Knee and Saturation Max. controls.

The Saturation Knee slider sets the image level at which saturation mapping begins. Below this level, no remapping is applied. All saturation values from this level on up are remapped according to the Saturation Max. slider. A value of 1.0 is maximum saturation in the currently selected output color space.

The Saturation Max. slider sets the new maximum level to which you want to either raise or lower all saturation values that are above the Saturation Knee setting. A value of 1.0 is maximum saturation in the currently selected output color space.

- **Clip:** Hard clips all out-of-gamut values.

NOTE: While this node has ACES settings, it does transforms to the ACES color space colorimetrically, which is not actually correct for ACES workflows. For actual ACES workflows, use the ACES Transform node, which uses transforms specified by the Academy.

Advanced

This drop-down menu exposes the advanced features of the Color Space Transform node.

- **Apply Forward OOTF:** Check this box to convert the image from scene referred to display referred color management.
- **Apply Inverse OOTF:** Check this box to convert the image from display referred to scene referred color management.
- **Use White Point Adaptation:** Applies a chromatic adaptation transform to account for different white points between color spaces.

Uncheck this box if you simply want to view the input color space's white point unaltered in the output color space. For example, wanting to use a P3-D60 mastered clip inside a P3-D65 timeline for reference purposes.

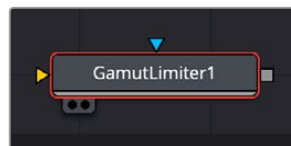
Check this box to apply the chromatic adaptation transform to convert the input white point to match the output color space's white point. For example, wanting a P3-D60 mastered clip to cut in with other clips mastered in a P3-D65 timeline.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Gamut Limiter [GMI]



The Gamut Limiter node

Gamut Limiter Node Introduction

Lets you limit the gamut to a specified standard. Useful in situations where the delivery color space is a large gamut such as Rec. 2020, but the QC specification requires limiting to a smaller gamut such as P3, in order to limit the amount of saturation allowable in the final output. This is a limiting operation, so out of bounds values are hard clipped. Because it's a limiter, it should probably be one of the last operations in any node tree to prevent useful image data from being clipped.

Inputs

The two inputs on the Gamut Limiter node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

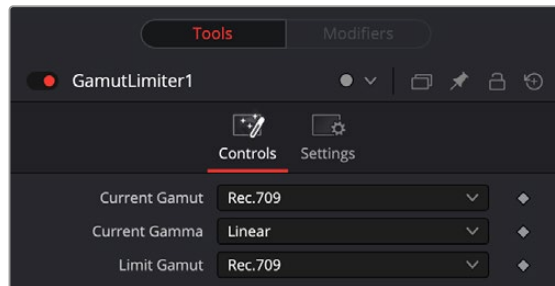
Basic Node Setup

The Gamut Limiter node, like many 2D image-processing nodes, receives a 2D image, such as a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Gamut Limiter node applied to a MediaIn1 node

Inspector



Gamut Limiter controls

Controls Tab

The Controls tab contains the primary controls necessary for adjusting the Gamut Limiter parameters.

Current Gamut

Choose the timeline gamut currently being used by the image.

Current Gamma

Choose the timeline gamma currently being used by the image.

Limit Gamut

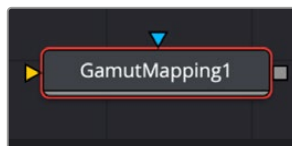
Choose the gamut you want to restrict the image to here.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Gamut Mapping [GMp]



The Gamut Mapping node

Gamut Mapping Node Introduction

The Color Space Transform node provides Gamut Mapping controls to accommodate workflows where you need to transform one color space into another that has a dramatically larger or smaller gamut. These controls are identical to those found in the Color Space Transform node's Gamut Mapping group and are similar to those found in the Color Management panel of the Project Settings.

Inputs

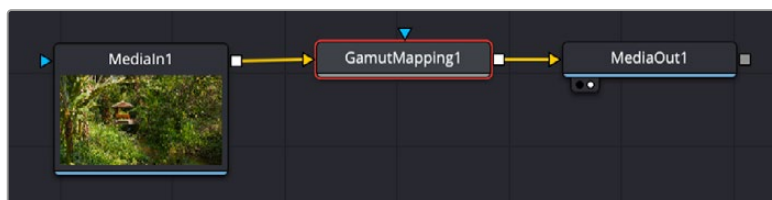
The two inputs on the Gamut Mapping node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

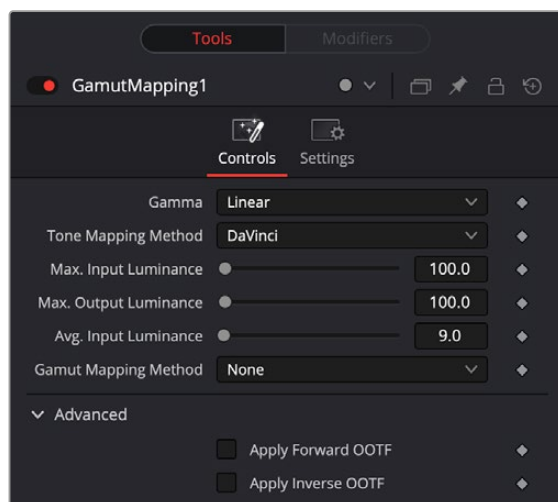
Basic Node Setup

The Gamut Mapping node, like many 2D image-processing nodes, receives a 2D image, such as a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Gamut Map node applied to a MediaIn1 node

Inspector



Gamut Mapping controls

Controls Tab

The Controls tab contains the primary controls necessary for adjusting the Gamut Limiter parameters.

Gamma

A pop-up menu lets you specify what type of gamma the clip is supposed to have, so set this to whatever matches that image (this may match the Timeline Color Space, but it depends on how you're working).

Tone Mapping Method

Lets you enable tone mapping to accommodate workflows where you need to transform one color space into another with a dramatically larger or smaller dynamic range, by automating an expansion or contraction of image contrast in such a way as to give a pleasing result with no clipping.

- **None:** This setting disables Input DRT Tone Mapping. No tone mapping is applied to the Input to Timeline Color Space conversion at all, resulting in a simple 1:1 mapping to the Timeline Color Space.
- **Clip:** Hard clips all out-of-bounds values.
- **Simple:** Uses a simple curve to perform this transformation, compressing or expanding the highlights and/or shadows of the timeline dynamic range to better fit the output dynamic range. Note that the "Simple" option maps between approximately 5500 nits and 100 nits, so if you're mapping from an HDR source with more than 5500 nits to an SDR destination there may still be some clipping of the highlights above 5500 nits.
- **Luminance Mapping:** Same as DaVinci, but more accurate when the Input Color Space of all your media is in a single standards-based color space, such as Rec. 709 or Rec. 2020.
- **DaVinci:** This option tone maps the transform with a smooth luminance roll-off in the shadows and highlights, and controlled desaturation of image values in the very brightest and darkest parts of the image. This setting is particularly useful for wide-gamut camera media and is a good setting to use when mixing media from different cameras.

- **Saturation Preserving:** This option has a smooth luminance roll-off in the shadows and highlights but does so without desaturating dark shadows and bright highlights, so this is an effective option for colorists who like to push color harder. However, because over-saturation in the highlights of the image can look unnatural, two parameters are exposed to provide some user-adjustable automated desaturation.

Sat. Rolloff Start: Lets you set a threshold, in nits (cd/m²), at which saturation will roll off along with highlight luminance. Beginning of the rolloff.

Sat. Rolloff Limit: Lets you set a threshold, in nits (cd/m²), at which the image will be totally desaturated. End of the rolloff.

Max Input/Output Luminance

Adjusting the sliders allows you to specify the minimum and maximum luminance of the input image in nits. Using these two sliders together, you can set which value from the Input Gamma is mapped to which value of the Output Gamma.

Average Input Luminance

Used to compensate for large differences in the viewer's state of visual adaptation when viewing a bright image on an HDR display versus seeing that same image on an SDR display. For most "average" images this setting works best set between 0–10. However, when you're converting very bright images (for example, a snow scene at noon), then using a higher value will yield more image detail within the highlights.

Gamut Mapping Method

Accommodates workflows where you need to transform one color space into another with a dramatically larger or smaller gamut by helping to automate an expansion or contraction of image saturation in such a way as to give a pleasing and naturalistic result with no clipping. Choosing Saturation Mapping lets you remap the saturation values of the image. It enables the Saturation Knee and Saturation Max. controls.

- The Saturation Knee slider sets the image level at which saturation mapping begins. Below this level, no remapping is applied. All saturation values from this level on up are remapped according to the Saturation Max. slider. A value of 1.0 is maximum saturation in the currently selected output color space.
- The Saturation Max. slider sets the new maximum level to which you want to either raise or lower all saturation values that are above the Saturation Knee setting. A value of 1.0 is maximum saturation in the currently selected output color space.

Advanced

- **Apply Forward OOTF:** Check this box to convert the image from scene referred to display referred color management.
- **Apply Inverse OOTF:** Check this box to convert the image from display referred to scene referred color management.

Common Controls

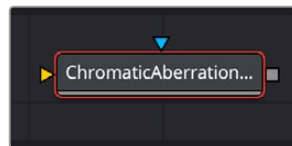
Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Revival

A sub category of the Color tools in Fusion that pertains to fixing common technical errors with programs being finished, remastered, or restored.

Chromatic Aberration Removal [CAr]



The Chromatic Aberration Removal node

Chromatic Aberration Removal Node Introduction

A node that lets you manually correct the slight color fringing that results from chromatic aberration in a lens. Show Estimated Fringes checkboxes for the RGB channels display an “alignment guide” that visually isolates each of the types of fringing against gray.

Inputs

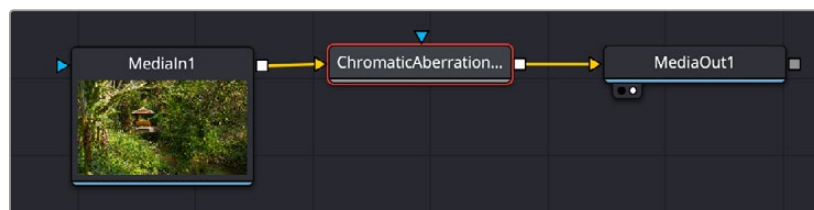
The two inputs on the Chromatic Aberration Removal node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

Basic Node Setup

The Chromatic Aberration Removal node, like many 2D image-processing nodes, receives a 2D image, such as a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Chromatic Aberration Removal node applied to a MediaIn1 node

Inspector



Chromatic Aberration Removal controls

Controls Tab

The Controls tab contains the primary controls necessary for adjusting the Chromatic Aberration Removal parameters.

Advanced Options

Advanced Options provide additional parameters for problem shots.

- **Lens Center:** Center X and Y parameters let you offset the center of the lens if you're dealing with a reframed and re-rendered shot.
- **Stronger Correction:** Shows the location of image features that resemble fringing.

Estimation Options

The Estimation Options are used solely to highlight and identify areas of fringing. They do not affect the final output of the image. They only become available when one of the Show Estimated Fringes boxes is checked in the Aberration Correction section.

- **R/C, G/P, B/Y Balance:** These tools adjust the incoming balance between their respective colors to better identify hard to see fringing.
- **Brightness:** Magnifies the fringe indicators that are displayed when you turn on either of the Estimated Fringes checkboxes.

Aberration Correction

These tools let you make manual adjustments to correct aberration issues.

- **R/C, G/P, B/Y Scale:** Adjust these sliders to eliminate the fringing from their respective colors.
- **R/C, G/P, B/Y Edge:** Adjusts to compensate for the difference in fringing due to the curvature of the lens' edges
- **Show Estimated Fringes:** Checking this box will show just the estimated fringes over a gray background. It will also activate the Estimation Options sliders, which will let you further highlight and identify areas of fringing.

This makes it easy to make manual adjustments to correct the problem, using the Scale and Edge controls to individually adjust Red/Cyan, Green/Purple, and Blue/Yellow fringing.

Common Controls

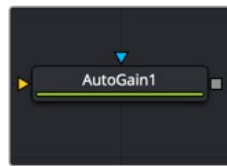
Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Color

A sub category of the Color tools in Fusion that gives you access to various color processing tools.

Auto Gain [AG]



The Auto Gain node

Auto Gain Node Introduction

The Auto Gain node automatically adjusts the tonal range of an image, setting the darkest and brightest pixels to user-selected values. By default, the darkest pixels get pushed to black, the brightest pixels get pushed to white, and pixels in between get stretched to cover the tonal range evenly.

This can be useful when compensating for variations in lighting, dealing with low-contrast images, or visualizing the full color range of float images (although the viewer's View Normalized Image option is generally more suitable for this).

Inputs

The two inputs on the Auto Gain node are the input and effect mask.

Input: The orange input connects the primary 2D image for the auto gain.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the auto gain adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

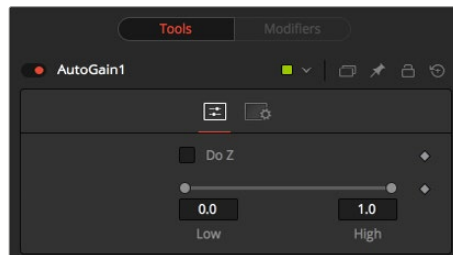
Basic Node Setup

The Auto Gain node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



An Auto Gain node applied to a MediaIn1 node

Inspector



Auto Gain controls

Controls Tab

The Controls tab contains the few primary controls necessary for customizing the AutoGain operation.

NOTE: Variations over time in the input image can cause corresponding variations in the levels of the result. For example, if a bright object moves out of an otherwise dark shot, the remaining scene gets suddenly brighter, as the remaining darker values get stretched to white. This also applies to sudden depth changes when Do Z is applied; existing objects may be pushed forward or backward when a near or far object enters or leaves the scene.

Do Z

Select the Do Z checkbox to apply the Auto Gain effect to the Z or Depth channels. This can be useful for matching the ranges of one Z-channel to another, or to view a float Z-channel in the RGB values.

Range

This Range control sets the black point and white point in the image. All tonal values in the image rescale to fit within this range.

EXAMPLE Create a horizontal gradient with the Background node. Set one color to dark gray (RGB Values 0.2). Set the other color to light gray (RGB Values 0.8).

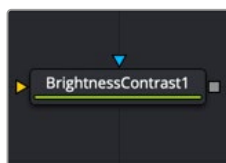
Add an Auto Gain node and set the Low value to 0.0 and the High value to 0.5. This causes the brightest pixels to be pushed down to 0.5, and the darkest pixels get pushed to black. The remainder of the pixel values scale between those limits.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Brightness Contrast [BC]



The Brightness Contrast node

Brightness Contrast Node Introduction

The Brightness Contrast node adjusts the gain, brightness, contrast, gamma, and saturation of an image. The order of the controls represents the order in which the operations are applied. For example, gamma gets applied before contrast but after gain. The Brightness Contrast is also reversible using the Forward and Reverse buttons. So color corrections, once applied, can be reversed further downstream.

For this to work best, image processing should operate in 32-bit floating point.

Inputs

The two inputs on the Brightness Contrast node are the input and effect mask.

Input: The orange input connects the primary 2D image for the brightness contrast.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Brightness Contrast adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

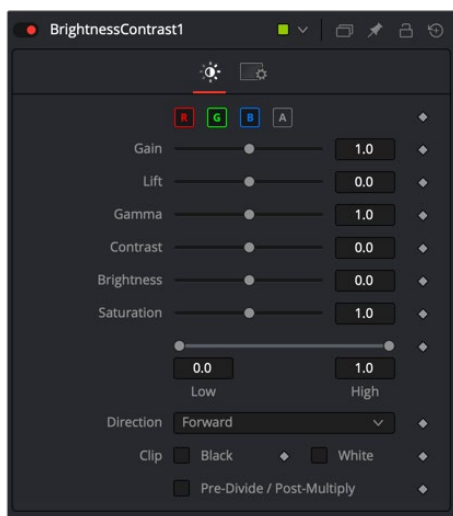
Basic Node Setup

The Brightness Contrast node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Brightness Contrast node applied to a MediaIn1 node

Inspector



Brightness Control controls

Controls Tab

The Controls tab contains all the primary controls necessary for customizing the brightness, contrast operations.

Color Channels (RGBA)

The filter defaults to operating on R, G, B, and A channels. Selective channel filtering is possible by clicking each channel button to make them active or inactive.

NOTE: This is not the same as the RGBA checkboxes found under the common controls. The node takes these selections into account before it processes the image, so deselecting a channel causes the node to skip that channel when processing, speeding up the rendering of the effect. In contrast, the channel controls under the Common Controls tab get applied after the node has processed.

Gain

The gain slider is a multiplier of the pixel value. A Gain of 1.2 makes a pixel that is R0.5 G0.5 B0.4 into R0.6 G0.6, B0.48 (i.e., $0.4 * 1.2 = 0.48$) while leaving black pixels unaffected. Gain affects higher values more than it affects lower values, so the effect is most influential in the midrange and top range of the image.

Lift

While Gain scales the color values around black, Lift scales the color values around white. The pixel values get multiplied by the value of this control. A Lift of 0.5 makes a pixel that is R0.0 G0.0 B0.0 into R0.5 G0.5, B0.5 while leaving white pixels unaffected. Lift affects lower values more than it affects higher values, so the effect is most influential in the midrange and low range of the image.

Gamma

Values higher than 1.0 raise the Gamma (mid-gray), whereas lower values decrease it. The effect of this node is not linear, and existing black or white points are not affected at all. Pure gray colors are affected the most.

Contrast

Contrast is the range of difference between the light to dark areas. Increasing the value of this slider increases the contrast, pushing color from the midrange toward black and white. Reducing the contrast causes the colors in the image to move toward midrange, reducing the difference between the darkest and brightest pixels in the image.

Brightness

The value of the Brightness slider gets added to the value of each pixel in the image. This control's effect on an image is linear, so the effect is applied identically to all pixels regardless of value.

Saturation

Use this control to increase or decrease the amount of Saturation in the image. A saturation of 0 has no color, reducing the image to grayscale.

Low and High

This range control is similar to the Gain control in some respects. If Low gets anchored at 0.0 and the High value gets reduced from 1.0, the effect is identical to increasing the gain. High values get multiplied by the inverse of the high value. (e.g., if high is 0.75, each pixel is multiplied by $1/0.75$, or 1.3333).

Leaving the high anchored at 1.0 and increasing the low is the same as inverting the image colors and increasing the gain and inverting it back again. This pushes more of the image toward black without affecting the whites at all.

Direction

Forward applies all values normally. Reverse effectively inverts all values.

Clip Black/White

The Clip Black and Clip White checkboxes clip out-of-range color values that can appear in an image when processing in floating-point color depth. Out-of-range colors are below black (0.0) or above white (1.0). These checkboxes have no effect on images processed at 8-bit or 16-bit per channel, as such images cannot have out-of-range values.

Pre-Divide/Post-Multiply

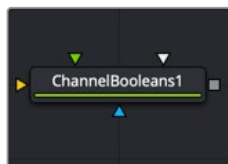
Selecting the Pre-Divide/Post-Multiply checkbox causes the image pixel values to be divided by the Alpha values before the color correction, and then re-multiplied by the Alpha value after the correction.

This helps to prevent the creation of illegally additive images when color correcting images with premultiplied Alpha channels.

Common Controls**Settings Tab**

The Settings tab in the Inspector appears in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Channel Booleans [Bol]



The Channel Booleans node

Channel Booleans Node Introduction

The Channel Booleans node applies a variety of mathematical and logical operations on the channels in an image. This node works by using one image's channels to modify another image's channels. If a foreground input is not available, selecting options that use color channels from the foreground ends up using the background input's color channels instead.

NOTE: Be aware of another similarly named Channel Boolean (3Bol), which is a 3D node used to remap and modify channels of 3D materials. When modifying 2D channels, use the Channel Booleans (with an "s") node (Bol).

Inputs

There are four inputs on the Channel Booleans node in the Node Editor, but only the orange Background input is required.

Background: This orange input connects a 2D image that gets adjusted by the foreground input image.

Effect Mask: The blue effect mask input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the channel booleans adjustment to only those pixels within the mask.

Foreground: The green foreground input connects a 2D image that is used to adjust the background input image.

Matte: The white matte input can be used to combine external mattes with the foreground and background operations.

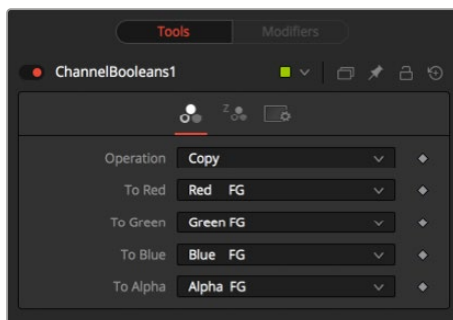
Basic Node Setup

The Channel Booleans node is an extremely flexible tool used in many different ways. The example below copies the z-depth channel from the foreground input (green) into the background image (orange).



A Channel Booleans set to copy from foreground to background

Inspector



Channel Booleans controls

Color Channel Tab

On the Color Channels tab, the controls are divided into two columns.

On the left side are target channels for the image connected into the orange background input. The drop-down menu to the right lets you choose whether you want to modify the BG image with its channels (suffix BG after list name) or with the channels from an image connected into the green foreground input on the node (suffix FG in the drop-down list).

Operation

This menu selects the mathematical operation applied to the selected channels. The options are as follows:

Copy

Copy the value from one color channel to another. For example, copy the foreground red channel into the background's Alpha channel to create a matte.

- **Add:** Add the color values from one color channel to another channel.
- **Subtract:** Subtract the color values of one color channel from another color channel.
- **And:** Perform a logical AND on the color values from color channel to color channel. The foreground image generally removes bits from the color channel of the background image.
- **Or:** Perform a logical OR on the color values from color channel to color channel. The foreground image generally adds bits from the color channel of the background image.
- **Exclusive Or:** Perform a logical XOR on the color values from color channel to color channel. The foreground image generally flips bits from the color channel of the background image.
- **Multiply:** Multiply the values of a color channel. This gives the appearance of darkening the image as the values scale from 0 to 1. White has a value of 1, so the result would be the same. Gray has a value of 0.5, so the result would be a darker image or, in other words, an image half as bright.
- **Divide:** Divide the values of a color channel. This gives the appearance of lightening the image as the values scale from 0 to 1.
- **Maximum:** Compare the two images and take the maximum, or brightest, values from each image.
- **Minimum:** Compare the two images and take the minimum, or darkest, values from each image.
- **Negative:** Invert the FG input to make a negative version of the image.
- **Solid:** Solid sets a channel to a full value of 255. This is useful for setting the Alpha to full value.
- **Clear:** Clear sets a channel to a value of zero. This is useful for clearing the Alpha.

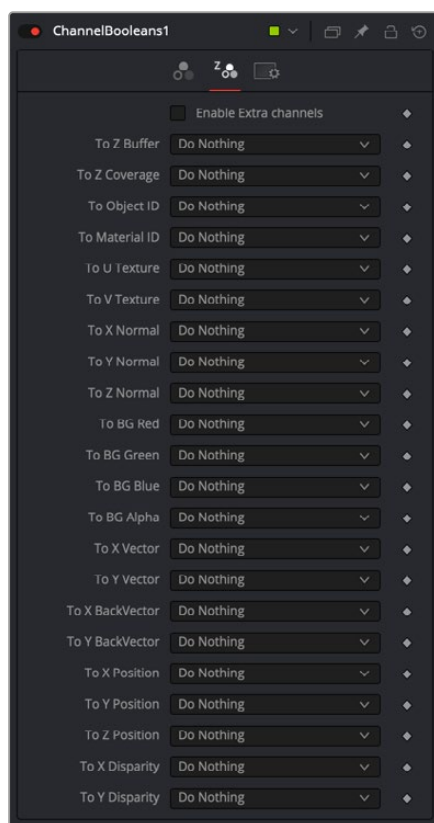
- **Difference:** Difference subtracts the greater color values of one color channel from the lesser values of another color channel.
- **Signed Add:** Signed Add subtracts areas that are lower than mid-gray and adds areas that are higher than mid-gray, which is useful for creating effects with embossed gray images.

To Red, To Green, To Blue, To Alpha

These menus represent the four color channels of the output image. Use the drop-down menu to select which channel from the source images produces the output channel.

The default setting copies the channels from the foreground channel. Select any one of the four color channels, as well as several auxiliary channels like Z-buffer, saturation, luminance, and hue.

Inspector



Aux Channel Inspector

Aux Channel Tab

This tab includes a series of menus where you select a source for the auxiliary channels of the output image.

Enable Extra Channels

When the Enable Extra Channels checkbox is selected, the Channel Booleans node can output images with channels beyond the usual RGBA. Once enabled, the remaining controls in the Aux Channels tab can copy data into the auxiliary channels.

EXAMPLES To copy the Alpha channel of one image to its color channels, set the red, green, and blue channels to Alpha BG. Set the Operation to Copy.

To copy the Alpha channel from another image, set operation type to Alpha FG.

To replace the existing Alpha channel of an image with the Alpha of another image, choose “Do Nothing” for To Red, To Green, and To Blue and “Alpha FG” for To Alpha. Pipe the image containing the Alpha into the foreground input on the Channel Booleans node. Set Operation: “Copy.” The same operation is available in the Matte Control node.

To combine any mask into an Alpha channel of an image, choose “Do Nothing” for To Red, To Green, and To Blue and “Matte” for To Alpha. Pipe the mask into the foreground input on the Channel Booleans node. Set Operation: “Copy.”

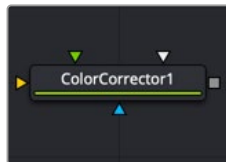
To subtract the red channel’s pixels of another image from the blue channel, choose “Do Nothing” for To Red and To Green and “Red FG” for To Blue. Pipe the image containing the red channel to subtract into the foreground input on the Channel Booleans node. Set Operation: “Subtract.”

Common Controls

Settings Tab

The Settings tab in the Inspector appears in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Color Corrector [CC]



The Color Corrector node

Color Corrector Node Introduction

The Color Corrector node is a comprehensive color node with histogram matching, and equalization, hue shifting, tinting, and color suppression.

Controls in the Color Corrector node are separated into four tabs: Correction, Ranges, Options, and Settings.

Inputs

The Color Corrector node includes four inputs in the Node Editor.

Input: This orange input is the only required connection. It connects a 2D image for color correction.

Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color corrector adjustment to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

Match Reference: The green input is used to connect an image that can be a reference for histogram matching.

Match Mask: This optional white input accepts any mask much like an effect mask. However, this mask defines of the area to match during a Histogram Match. It offers more flexibility in terms of shape than the built-in Match rectangle in the Inspector.

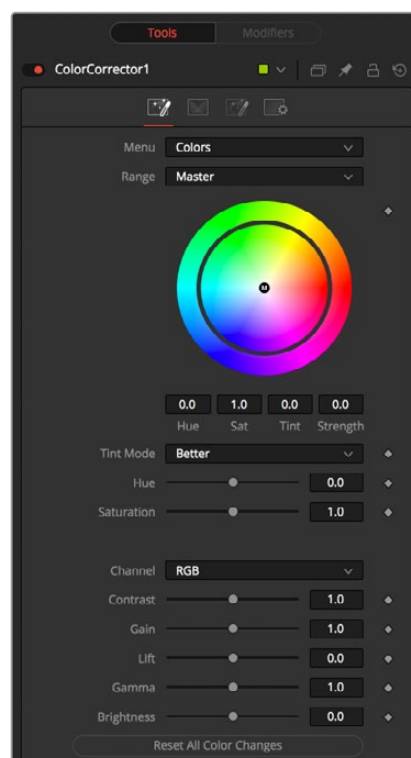
Basic Node Setup

The Color Corrector node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Color Corrector node applied to a MediaIn1 node

Inspector



Color Corrector controls

Correction Tab Colors Menu

The main Correction tab is further separated into four types of correction methods: colors, levels, histogram, and suppress. Selecting one from the menu at the top of the Correction tab causes that method's controls to appear. The Color method is described in detail below.

Range

This menu determines the tonal range affected by the color correction controls in this tab. The menu can be set to Shadows, Midtones, Highlights, and Master, where Master is the default affecting the entire image.

The selected range is maintained throughout the Colors, Levels, and Suppress sections of the Color Corrector node.

Adjustments made to the image in the Master channel are applied to the image after any changes made to the Highlight, Midtone, and Shadow ranges.

NOTE: The controls are independent for each color range. For example, adjusting the Gamma control while in Shadows mode does not change or affect the value of the Gamma control for the Highlights mode. Each control is independent and applied separately.

Color Wheel

The color wheel provides a visual representation of adjustments made to Hue and Saturation, as well as any tinting applied to the image. Adjustments can be made directly by dragging the color indicator, or by entering values in the numeric boxes under the color wheel.

The tinting is represented in the color wheel color indicator that shows the color and strength of the tint. The Highlight setting uses a black outline for the color indicator. The Midtones and Shadows use gray color indicators. The Master color indicator is also black, but it has a white M in the center to distinguish it from the others.

The mouse can position the color indicator for each range only when the applicable range is selected. For example, the Highlight color indicator cannot be moved when the Master range is selected.

Holding down the Command or Ctrl key while dragging this indicator allows you to make finer adjustments by reducing the control's sensitivity to mouse movements. Holding down the Shift key limits the movement of the color indicator to a single axis, allowing you to restrict the effect to either tint or strength.

Tint Mode

This menu is used to select the speed and quality of the algorithm used to apply the hue and saturation adjustments. The default is Better, but for working with larger images, it may be desirable to use a faster method.

Hue

This slider is a clone of the Hue control located under the color wheel. The slider makes it easier to make small adjustments to the value with the mouse. The Hue control provides a method of shifting the hue of the image (or selected color range) through the color spectrum. The control value has an effective range between -0.1 and 1.0, which represents the angle of rotation in a clockwise direction. A value of 0.25 would be 90 degrees (90/360) and would have the effect of shifting red toward blue, green to red, and so on.

Hue shifting can be done by dragging the slider, entering a value directly into the text control, or by placing the mouse above the outer ring of the color wheel and dragging the mouse up or down. The outer ring always shows the shifted colors compared to the original colors shown in the center of the wheel.

Saturation

This slider is a clone of the Saturation control located under the color wheel. The slider makes it easier to make small adjustments to the value with the mouse. The Saturation control is used to adjust the intensity of the color values. A saturation of 0 produces gray pixels without any color component, whereas a value of 1.0 produces no change in the chroma component of the input image. Higher values generate oversaturated values with a high color component.

Saturation values can be set by dragging the slider, entering a value directly into the text control, or by dragging the mouse to the left and right on the outer ring of the color wheel control.

Channel

This menu is set for the Histogram, Color, and Levels sections of the Color Corrector node. When the red channel is selected, the controls in each mode affect the red channel only, and so on.

The controls are independent, so switching to blue does not remove or eliminate any changes made to red, green, or Master. The animation and adjustments made to each channel are separate. This menu simply determines what controls to display.

Contrast

Contrast is the range of difference between the light to dark areas. Increasing the value of this slider increases the contrast, pushing color from the midrange toward black and white. Reducing the contrast causes the colors in the image to move toward midrange, reducing the difference between the darkest and brightest pixels in the image.

Gain

The Gain slider is a multiplier of the pixel value. A gain of 1.2 makes a pixel that is R0.5 G0.5 B0.4 into R0.6 G0.6, B0.48 (i.e., $0.4 * 1.2 = 0.48$), while leaving black pixels totally unaffected. Gain affects higher values more than it affects lower values, so the effect is strongest in the midrange and top range of the image.

Lift

While Gain scales the color values around black, Lift scales the color values around white. The pixel values are multiplied by the value of this control. A Lift of 0.5 makes a pixel that is R0.0 G0.0 B0.0 into R0.5 G0.5, B0.5, while leaving white pixels totally unaffected. Lift affects lower values more than it affects higher values, so the effect is strongest in the midrange and low range of the image.

Gamma

Values higher than 1.0 raise the Gamma (mid gray), whereas lower values decrease it. The effect of this node is not linear, and existing black or white points are not affected at all. Pure gray colors are affected the most.

Brightness

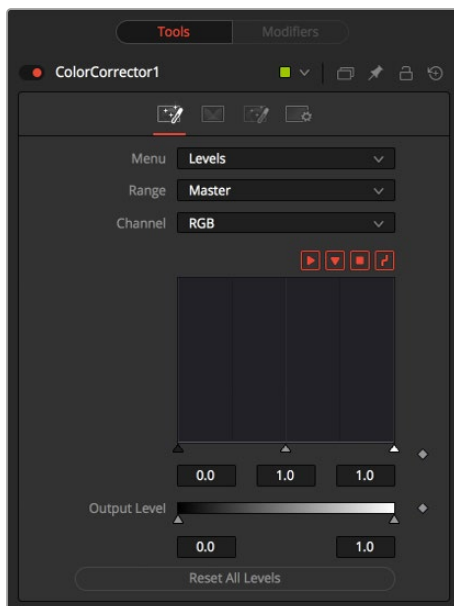
The value of the Brightness slider is added to the value of each pixel in your image. This control's effect on an image is linear, so the effect is applied identically to all pixels despite value.

Reset All Color Changes

Selecting this button returns all color controls in this section to their default values.

Correction Tab Levels Menu

The main Correction tab is further separated into four types of correction methods: colors, levels, histogram, and suppress. When Levels is selected from the menu, you can remap the white and black points of an image, with a Gamma control to adjust midtones. A histogram provides a view of the tonal distribution in the image to help guide your adjustments. The Level method is described in detail below.



Color Corrector Levels controls

Range

Identical to the Range menu when Color is selected in the Menu, the Range menu determines the tonal range affected by the color correction controls in this tab. The menu can be set to Shadows, Midtones, Highlights, and Master, where Master is the default affecting the entire image.

The selected range is maintained throughout the Colors, Levels, and Suppress sections of the Color Corrector node.

Adjustments made to the image in the Master channel are applied to the image after any changes made to the Highlights, Midtones, and Shadows ranges.

NOTE: The controls are independent for each color range. For example, adjusting the Gamma control while in Shadows mode does not change or affect the value of the Gamma control for the Highlights mode. Each control is independent and applied separately.

Channel

This menu is used to select and display the histogram for each color channel or for the Master channel.

Histogram Display

A histogram is a chart that represents the distribution of color values in the scene. The chart reads from left to right, with the leftmost values representing the darkest colors in the scene and the rightmost values representing the brightest. The more pixels in an image with the same or similar value, the higher that portion of the chart is.

Luminance is calculated per channel; therefore, the red, green, and blue channels all have their own histogram, and the combined result of these comprises the Master Histogram.

To scale the histogram vertically, place the mouse pointer inside the control and drag the pointer up to zoom in or down to zoom out.

Display Selector Toolbar

The Display Selector toolbar at the top of the histogram provides a method of enabling and disabling components of the histogram display. Hold the mouse pointer over the button to display a tooltip that describes the button's function.

- **Input Histogram:** This enables or disables the display of the input image's histogram.
- **Reference Histogram:** This enables or disables the display of the reference image's histogram.
- **Output Histogram:** This enables or disables the display of the histogram from the post-color-corrected image.
- **Corrective Curve:** This toggles the display of a spline used to visualize exactly how auto color corrections applied using a reference image are affecting the image. This can be useful when equalizing luminance between the input and reference images.

Histogram Controls

These controls along the bottom of the histogram display are used to adjust the input image's histogram, compressing or shifting the ranges of the selected color channel.

The controls can be adjusted by dragging the triangles beneath the histogram display to the left and right.

Shifting the High value toward the left (decreasing the value) causes the histogram to slant toward white, shifting the image distribution toward white. The Low value has a similar effect in the opposite direction, pushing the image distribution toward black.

Output Level

The Output Level control can apply clipping to the image, compressing the histogram. Decreasing the High control reduces the value of pixels in the image, sliding white pixels down toward gray and gray pixels toward black.

Adjusting the Low control toward High does the opposite, sliding the darkest pixels toward white.

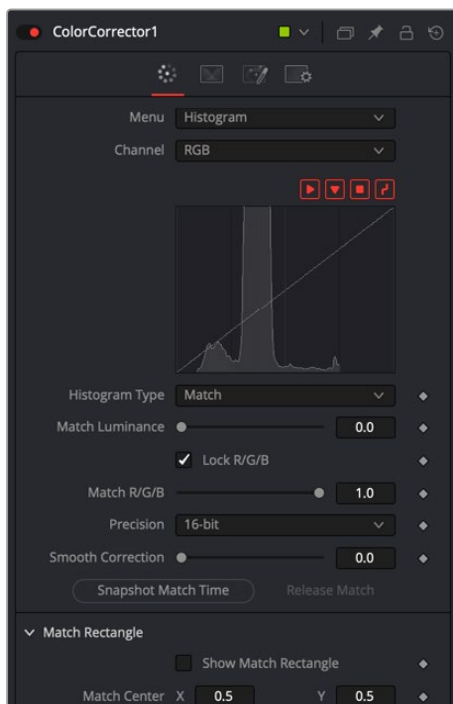
If the low value were set to 0.1, pixels with a value of 0.0 would be set to 0.1 instead, and other values would increase to accommodate the change. The best way to visualize the effect is to observe the change to the output histogram displayed above.

Reset All Levels

Clicking this button resets all the controls in the Levels section to their defaults.

Correction Tab Histogram Menu

When the menu is set to Histogram, a histogram display is produced of the input image. If a reference image is also provided, the histogram for the reference image is also displayed. The controls in this tab are primarily used to match one image to another, using either the Equalize or Match modes of the Color Corrector.



Color Correction in Histogram mode

Channel

This menu is used to select and display the histogram for each color channel or for the Master channel.

Histogram Display

A histogram is a chart that represents the distribution of color values in the scene. The chart reads from left to right, with the leftmost values representing the darkest colors in the scene and the rightmost values representing the brightest. The more pixels in an image with the same or similar value, the higher that portion of the chart is.

Luminance is calculated per channel; therefore, the red, green, and blue channels all have their own histogram, and the combined result of these comprises the Master Histogram.

To scale the histogram vertically, place the mouse pointer inside the control and drag the pointer up to zoom in or down to zoom out.

Display Selector Toolbar

The Display Selector toolbar at the top of the histogram provides a method of enabling and disabling components of the histogram display. Hold the mouse pointer over the button to display a tooltip that describes the button's function.

- **Input Histogram:** This enables or disables the display of the input image's histogram.
- **Reference Histogram:** This enables or disables the display of the reference image's histogram.
- **Output Histogram:** This enables or disables the display of the histogram from the post-color-corrected image.

- **Corrective Curve:** This toggles the display of a spline used to visualize exactly how auto color corrections applied using a reference image are affecting the image. This can be useful when equalizing luminance between the input and reference images.

Float Images and Histogram Equalization or Matching

By using the Histogram Match or Equalize methods on a float image, the color depth of the output image is converted to 16-bit integer. Two-dimensional histograms are not well suited to working with the extreme dynamic range of float images, so these operations always revert to 16-bit integer processing.

Histogram Type

Each of these menu options enables a different type of color correction operation.

- **Keep:** Keep produces no change to the image, and the reference histogram is ignored.
- **Equalize:** Selecting Equalize adjusts the source image so that all the color values in the image are equally represented—in essence, flattening the histogram so that the distribution of colors in the image becomes more even.
- **Match:** The Match mode modifies the source image based on the histogram from the reference image. It is used to match two shots with different lighting conditions and exposures so that they appear similar.

When selected, the Equalize and Match modes reveal the following controls.

- **Match/Equalize Luminance:** This slider affects the degree that the Color Corrector node attempts to affect the image based on its luminance distribution. When this control is zero (the default), matching and equalization are applied to each color channel independently, and the luminance, or combined value of the three color channels, is not affected.

If this control has a positive value when equalizing the image, the input image's luminance distribution is flattened before any color equalization is applied.

If this control has a positive value when the correction mode is set to Match, the luminance values of the input are matched to the reference before any correction is applied to the R, G, and B channels.

The Luminance and RGB controls can have a cumulative effect, and generally they are not both set to full (1.0) simultaneously.

- **Lock R/G/B:** When this checkbox is selected, color matching is applied to all color channels equally. When the checkbox is not selected, individual controls for each channel appear.

Equalize/Match R/G/B

The name of this control changes depending on whether the Equalize or Match modes have been selected. The slider can be used to reduce the correction applied to the image to equalize or match it. A value of 1.0 causes the full effect of the Equalize or Match to be applied, whereas lower values moderate the result.

Precision

This menu determines the color fidelity used when sampling the image to produce the histogram. 10-bit produces higher fidelity than 8-bit, and 16-bit produces higher fidelity than 10-bit.

Smooth Correction

Often, color equalization and matching operations introduce posterization in an image, which occurs because gradients in the image have been expanded or compressed so that the dynamic range between colors is not sufficient to display a smooth transition. This control can be used to smooth the correction curve, blending some of the original histogram back into the result for a more even transition.

Snapshot Match Time

Click this button to take a freeze snapshot of the current reference histogram, storing its current state as a snapshot in memory. If the reference histogram is not snapshot, the reference histogram is updated from frame to frame. This can cause flickering and phasing of the correction as the node tries to match a changing source to a changing reference.

Release Match

Click this button to release the current snapshot of the histogram and return to using the live reference input.

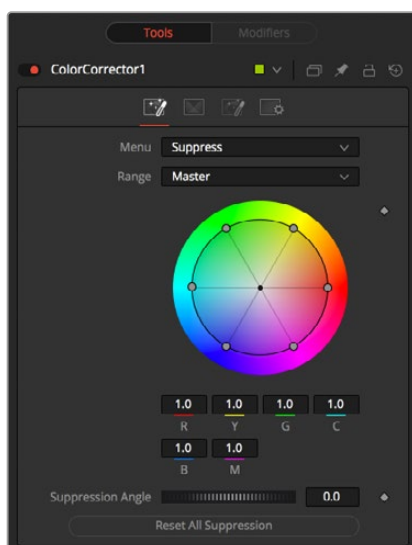
Reset All Histogram Changes

Selecting this button removes all changes made to the histogram, returning the controls to default and setting the mode back to Keep.

Correction Tab Suppress Menu

Color Suppression provides a mechanism for removing an unwanted color component from the image. The Color Wheel control is similar to that shown in the Colors section of the node, but this one is surrounded by six controls, each representing a specific color along the wheel.

To suppress a color in the selected range, drag the control that represents that color toward the center of the color wheel. The closer the control is to the center, the more that color is suppressed from the image.



Color Corrector Suppression controls

Suppression Angle

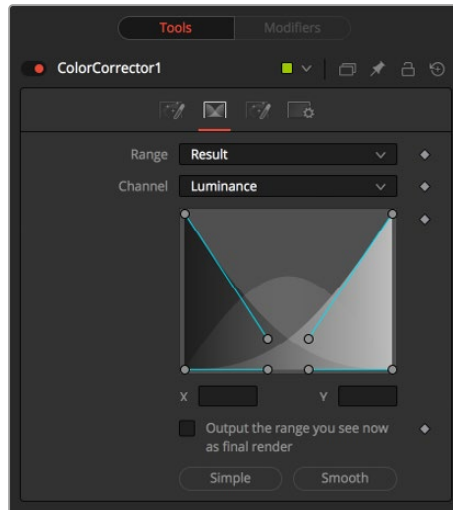
Use the Suppression Angle control to rotate the controls on the suppression wheel and zero in on a specific color.

Reset All Suppression

Clicking this control resets the suppression colors to 1.0, the default value.

Ranges Tab

The Ranges tab contains the controls used to specify which pixels in an image are considered to be shadows and which are considered to be highlights. The midrange is always calculated as pixels not already included in the shadows or the highlights.



Color Corrector Luminance controls

Range

This menu is used to select the tonal range displayed in the viewers. They help to visualize the pixels in the range. When the Result menu option is selected, the image displayed by the color corrector in the viewers is that of the color corrected image. This is the default.

Selecting one of the other menu options switches the display to a grayscale image showing which pixels are part of the selected range. White pixels represent pixels that are considered to be part of the range, and black pixels are not in the range. For example, choosing Shadows would show pixels considered to be shadows as white and pixels that are not shadows as black. Mid gray pixels are only partly in the range and do not receive the full effect of any color adjustments to that range.

Channel

The Channel menu in this tab can be used to examine the range of a specific color channel. By default, Fusion displays the luminance channel when the color ranges are examined.

Spline Display

The ranges are selected by manipulating the spline handles. There are four spline points, each with one Bézier handle. The two handles at the top represent the start of the shadow and highlight ranges, whereas the two at the bottom represent the end of the range. The Bézier handles are used to control the falloff.

The midtones range has no specific controls since its range is understood to be the space between the shadow and the highlight ranges.

The X and Y text controls below the spline display can be used to enter precise positions for the selected Bézier point or handle.

Output the Range You See Now as Final Render

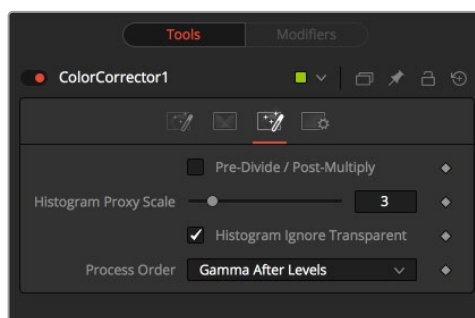
Selecting this checkbox causes the monochrome display of the range shown in the viewers to be output as the final render. Normally, the Color node outputs the full RGBA image, even if the node were left to display one of the color ranges in the view instead. This control makes it possible to use the Color Corrector node to generate a range's matte for use as an effect mask in other nodes.

Preset Simple/Smooth Ranges

These two buttons can be used to return the spline ranges to either Smooth (default) or Simple (linear) settings.

Options Tab

The Options tab includes a few very important processing operations including a simple solution when color correcting premultiplied Alpha channels.



Color Corrector controls

Pre-Divide/Post-Multiply

Selecting this option divides the color channels by the value of the Alpha before applying the color correction. After the color correction, the color values are re-multiplied by the Alpha to produce a properly additive image. This is crucial when performing an additive merge or when working with CG images generated with premultiplied Alpha channels.

Histogram Proxy Scale

The Histogram Proxy Scale determines the precision used when creating and calculating histograms. Lower values represent higher precision, and higher values produce a rougher, generalized histogram.

Process Order

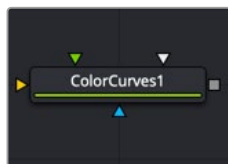
This menu is used to select whether adjustments to the image's gamma are applied before or after any changes made to the images levels.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Color Curves [CCv]



The Color Curves node

Color Curves Node Introduction

The Color Curves node is a spline-based node for performing Lookup table (LUT) color manipulations. A separate spline is provided for each color channel. The effect can be animated or dissolved and can be applied to the image using RGB, YUV, YIQ, CMY, or HLS color spaces.

The LUT view in the Color Corrector can be scaled using the + and - keys on the numeric keypad. The color curves LUT fully supports out-of-range values—i.e., pixels with color values above 1.0 or below 0.0.

The splines shown in this LUT view are also available from the Spline Editor, should greater precision be required when adjusting the controls.

Inputs

The Color Curves node includes three inputs in the Node Editor.

Input: This orange input is the only required connection. It connects a 2D image that is adjusted by the color curves.

Effect Mask: The optional effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color curves adjustment to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

Reference Image: The optional green input is used to connect a second 2D image that can be used for reference matching.

Match Mask: This optional white input accepts any mask much like an effect mask. However, this mask defines of the area to match during a Match. It offers more flexibility in terms of shape than the built-in Match reference rectangle in the Inspector.

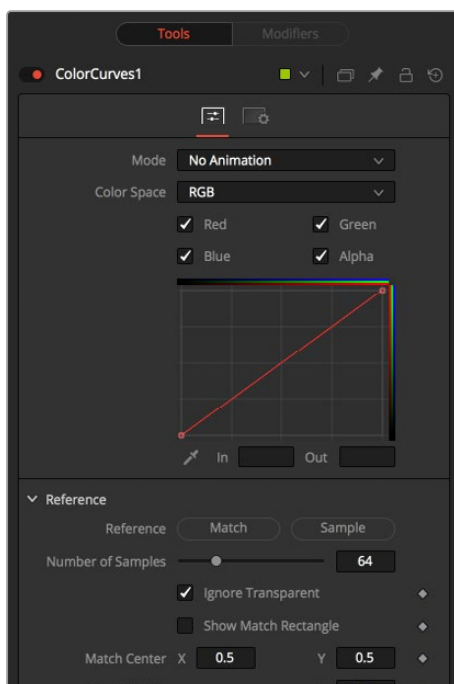
Basic Node Setup

The Color Curves node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Color Curves node applied to a MediaIn1 node

Inspector



Color Curves controls

Controls Tab

The Controls tab for the color curves is divided into two sections. The top half of the Inspector includes the curves and LUT controls. The bottom half is dedicated primarily to matching the reference image.

Mode

The Mode options change between Animated and Dissolve modes. The default mode is No Animation, where adjustments to the curves are static. Setting the mode provides a change spline for each channel, allowing the color curve to be animated over time.

Dissolve mode is essentially obsolete and is included for compatibility reasons only.

Color Space

The splines in the LUT view represent color channels from a variety of color spaces. The default is Red, Green, and Blue. The options in this menu allow an alternate color space to be selected.

A detailed description of the color spaces available here are below:

- **RGB (Red, Green, Blue):** Fusion uses the RGB color space, and most nodes and displays interpret the primary channels of an image as Red, Green, and Blue.
- **YUV (Luma, Blue Chroma, and Red Chroma):** The YUV color space is used in the analog broadcast of PAL video. Historically, this format was often used to color correct images, because of its familiarity to a large percentage of video engineers. Each pixel is described in terms of its Luminance, Blue Chroma, and Red Chroma components.
- **HLS (Hue, Luminance, and Saturation):** Each pixel in the HLS color space is described in terms of its Hue, Luminance, and Saturation components.

- **YIQ (Luma, In Phase, and Quadrature):** The YIQ color space is used in the analog broadcast of NTSC video. This format is much rarer than YUV and almost never seen in production. Each pixel is described in terms of its Luminance, Chroma (in-phase or red-cyan channel) and Quadrature (magenta-green) components.
- **CMY (Cyan, Magenta, and Yellow):** Although more common in print, the CMY format is often found in computer graphics from other software packages. Each pixel is described in terms of its Cyan, Magenta, and Yellow components. CMY is nonlinear.

Color Channels (RGBA)

Use the Color Channel controls to select which channel's spline is currently active for editing. The labels of these controls change to reflect the names of the channels for the current color space. Normally, they are read as Red, Green, and Blue. If the Color Curves node is operating in YUV color space, they are read as Y, U, and V instead.

These controls do not restrict the effect of the node to a specific channel. They only select whether the spline for that channel is editable. These controls are most often used to ensure that adding or moving points on one channel's spline do not unintentionally affect a different channel's spline.

Spline Window

The Spline Window displays a standard curve editor for each RGBA channel. These splines can be edited individually or as a group, depending on the color channels selected above.

The spline defaults to a linear range, from 0 in/0 out at the bottom left to the 1 in/1 out at the top right. At the default setting, a color processes to the same value as the output. If a point is added in the middle at 0.5 in/0.5 out, and the point is moved up, this raises the mid color of the image brighter.

The spline curves allow precise control over color ranges, so specific adjustments can be made without affecting other color values.

In and Out

Use the In and Out controls to manipulate the precise values of a selected point. To change a value, select a point and enter the in/out values desired.

Eyedropper (Pick)

Click the Eyedropper icon, also called the Pick button, and select a color from an image in the display to automatically set control points on the spline for the selected color. The new points are drawn with a triangular shape and can only be moved vertically (if point is locked, only the Out value can change).

Points are only added to enabled splines. To add points only on a specific channel, disable the other channels before making the selection.

One use for this technique is white balancing an image. Use the Pick control to select a pixel from the image that should be pure gray. Adjust the points that appear so that the Out value is 0.5 to change the pixel colors to gray.

Use the contextual menu's Locked Pick Points option to unlock points created using the Pick option, converting them into normal points.

Reference

The Reference section includes controls that handle matching to sample areas of the connected reference image.

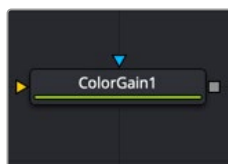
- **Match Reference:** The Match Reference button adds points on the curve to match an image connected to the green reference image input. The number of points used to match the image is based on the Number of Samples slider below.
- **Sample Reference:** Clicking the Sample Reference button samples the center scanline of the background image and creates a LUT of its color values. The number of points used to match the samples scanline is based on the Number of Samples slider below.
- **Number of Samples:** This slider determines how many points are used to match the curve to the range in the reference image.
- **Show Match Rectangle:** Enabling this checkbox displays a rectangle in the viewer showing the area on the reference image used during the match process. The match rectangle affects only the result of the Match Reference operation. The Sample reference is always done from the center scaling of the image.
- **Match Center:** The X and Y parameters allow you to reposition the match rectangle to sample a different area when matching.
- **Match Width:** Width controls the width of the match rectangle.
- **Match Height:** Height controls the height of the match rectangle.
- **Pre-Divide/Post-Multiply:** Selecting this checkbox causes the image's pixel values to be divided by the Alpha values prior to the color correction, and then re-multiplied by the Alpha value after the correction. This helps to avoid the creation of illegally additive images, particularly around the edges of a blue/green key or when working with 3D-rendered objects.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Color Gain [Clr]



The Color Gain node

Color Gain Node Introduction

The Color Gain node contains options for adjusting the gain, gamma, saturation, and hue of the image. Many controls provided by the Color Gain node are also found in the Color Corrector node, but this simpler node may render more quickly. One feature that distinguishes the Color Gain node from the Color Corrector is its balance tab controls. These can be used to adjust the tinting of the colors in the highs, mids, and lows.

Inputs

The Color Gain node includes two inputs: one for the main image and the other for an effect mask.

Input: This orange input is the only required connection. It connects a 2D image that gets adjusted by the color gain.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color gain adjustment to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

Basic Node Setup

The Color Gain node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D-image processing node or a Merge node.



A Color Gain node applied to a MediaIn1 node

Inspector



Color Gain controls

Gain Tab

The Gain tab provides control of individual RGBA Lift/Gamma/Gain parameters. These controls can quickly enable you to fix irregular color imbalances in specific channels.

Lock R/G/B

When selected, the Red, Green, and Blue channel controls for each effect are combined into one slider. Alpha channel effects remain separate.

Gain RGBA

The Gain RGBA controls multiply the values of the image channel in a linear fashion. All pixels are multiplied by the same factor, but the effect is larger on bright pixels and smaller on dark pixels. Black pixels do not change because multiplying any number times 0 is always 0.

Lift RGBA

While Gain scales the color values around black, Lift scales the color values around white. The pixel values are multiplied by the value of this control. A Lift of 0.5 makes a pixel that is R0.0 G0.0 B0.0 into R0.5 G0.5, B0.5, while leaving white pixels totally unaffected. Lift affects lower values more than it affects higher values, so the effect is strongest in the midrange and low range of the image.

Gamma RGBA

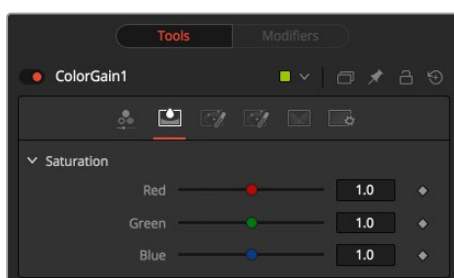
The Gamma RGBA controls affect the brightness of the midrange in the image. The effect of this node is nonlinear. White and black pixels in the image are not affected when gamma is modified, whereas pure grays are affected most by changes to this parameter. Large changes to this control tend to push midrange pixels into black or white, depending on the value used.

Pre-Divide/Post-Multiply

Selecting this checkbox causes the image pixel values to be divided by the Alpha values prior to the color correction, and then re-multiplied by the Alpha value after the correction. This helps when attempting to color correct images with premultiplied Alpha channels.

Saturation Tab

This Setting tab includes controls for the intensity of the colors in the individual RGB channels.



Color Gain Saturation setting tab

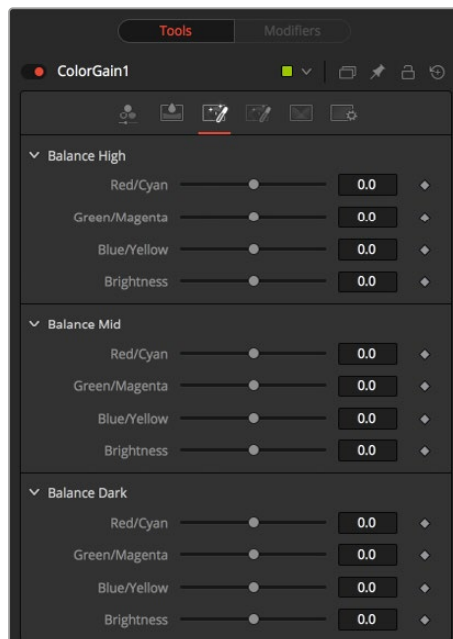
RGB Saturation

When adjusting an individual channel, a value of 0.0 strips out all that channel's color. Values greater than one intensify the color in the scene, pushing it toward the primary color.

Balance Tab

This tab in the Color Gain node offers controls for adjusting the overall balance of a color channel. Independent color and brightness controls are offered for the High, Mid, and Dark ranges of the image.

Colors are grouped into opposing pairs from the two dominant color spaces. Red values can be pushed toward Cyan, Green values toward Magenta and Blue toward Yellow. Brightness can be raised or lowered for each of the channels.



Color Gain Balance tab

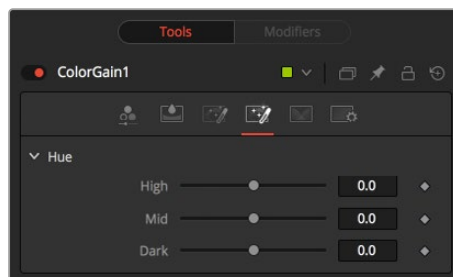
CMY Brightness Highs/Mids/Darks

By default, the Balance sliders can be adjusted by -1 to +1, but values outside this range can be entered manually to increase the effect. A value of 0.0 for any slider indicates no change to the image channel. Positive and negative values indicate that the balance of the image channel has been pushed toward one color or the other in the pair.

Hue Tab

Use the Hue tab of the Color Gain node to shift the overall hue of the image, without affecting the brightness, or saturation. Independent controls of the High, Mid, and Dark ranges are offered by three sliders.

The following is the order of the hues in the RGB color space: Red, Yellow, Green, Cyan, Blue, Magenta and Red.



Color Gain Hue tab

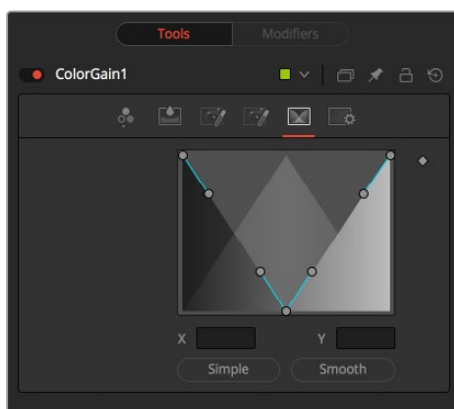
High/Mid/Dark Hue

Values above 0 push the hue of the image toward the right (red turns yellow). Values below 0 push the hue toward the left (red turns magenta). At -1.0 or 1.0, the hue completes the cycle and returns to its original value.

The default range of the hue sliders is -1.0 to +1.0. Values outside this range can be entered manually.

Ranges Tab

The Ranges tab contains the controls used to specify which pixels in an image are considered to be shadows and which are considered to be highlights. The midrange is always calculated as pixels not included in either the shadows or the highlights.



Color Gain Ranges tab

Spline Display

The ranges are selected by manipulating the spline handles. There are four spline points, each with one Bézier handle. The two handles at the top represent the start of the shadow and highlight ranges, whereas the two at the bottom represent the end of the range. The Bézier handles are used to control the falloff.

The midtones range has no specific controls since its range is understood to be the space between the shadow and the highlight ranges. The X and Y text controls below the Spline display can be used to enter precise positions for the selected Bézier point or handle.

Preset Simple/Smooth Ranges

These two buttons can be used to return the spline ranges to either Smooth (default) or Simple (linear) settings.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Color Matrix [CMx]



The Color Matrix node

Color Matrix Node Introduction

The ColorMatrix allows a vast number of operations to modify values individually in the different color channels.

Inputs

The Color Matrix node includes two inputs: one for the main image and the other for an effect mask.

Input: This orange input is the only required connection. It connects a 2D image that is adjusted by the color matrix.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color matrix adjustment to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

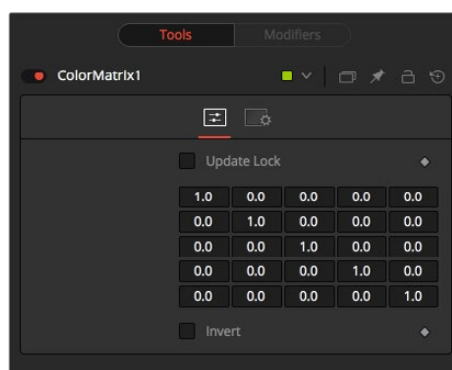
Basic Node Setup

The Color Matrix node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Color Matrix node applied to a MediaIn1 node

Inspector



Color Matrix controls

Controls Tab

Color Matrix multiplies the RGBA channels based on the values entered in a 4 x 4 grid. The fifth column/row is an Add column.

Update Lock

When this control is selected, Fusion does not render the node. This is useful for setting up each value of the node, and then turning off Update Lock to render it.

Matrix

This defines what type of operation actually takes place. The horizontal rows define the output values of the node. From left to right, they are R, G, B, A, and Add. The vertical columns define the input values. From top to bottom, they are R, G, B, A, and Add. The Add column allows simple adding of values to the individual color channels.

By default, the output values are identical to the input values.

- 1.0 means 100% of the Red channel input is copied to the Red channel output.
- 1.0 means 100% of the Green channel input is copied to the Green channel output.
- 1.0 means 100% of the Blue channel input is copied to the Blue channel output.
- 1.0 means 100% of the Alpha channel input is copied to the Alpha channel output.

Written as mathematical equations, the default settings of the matrix would appear as follows:

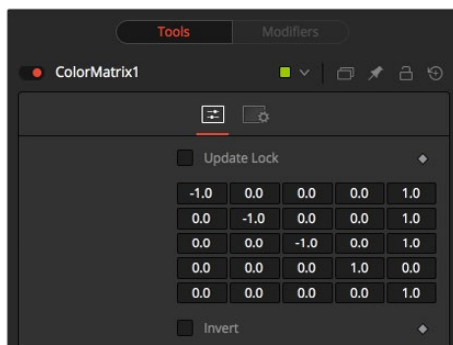
$$\begin{aligned} [R \text{ out}] &= 1 * [R \text{ in}] + 0 * [G \text{ in}] + 0 * [B \text{ in}] + 0 * [A \text{ in}] + 0 \\ [G \text{ out}] &= 0 * [R \text{ in}] + 1 * [G \text{ in}] + 0 * [B \text{ in}] + 0 * [A \text{ in}] + 0 \\ [B \text{ out}] &= 0 * [R \text{ in}] + 0 * [G \text{ in}] + 1 * [B \text{ in}] + 0 * [A \text{ in}] + 0 \\ [A \text{ out}] &= 0 * [R \text{ in}] + 0 * [G \text{ in}] + 0 * [B \text{ in}] + 1 * [A \text{ in}] + 0 \end{aligned}$$

Invert

Enabling this option inverts the Matrix. Think of swapping channels around, doing other operations with different nodes, and then copying and pasting the original ColorMatrix and setting it to Invert to get your channels back to the original.

Example 1: Invert

If you want to do a simple invert or negative of the color values, but leave the Alpha channel untouched, the matrix would look like this:

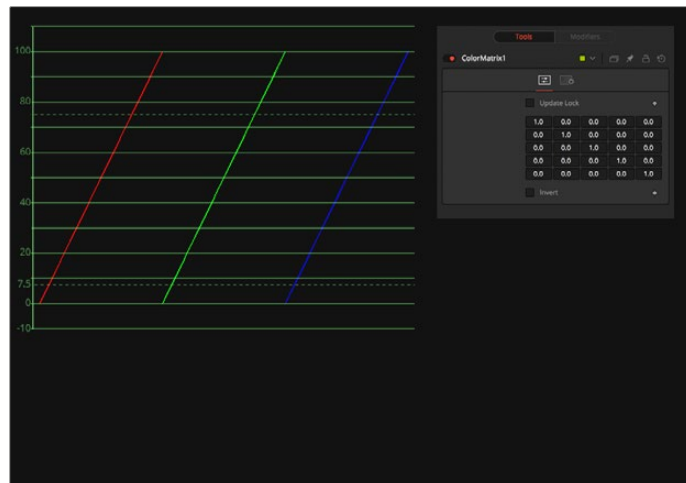


Color Matrix example

Observe the fact that we have to add 1 to each channel to push the inverted values back into the positive numbers.

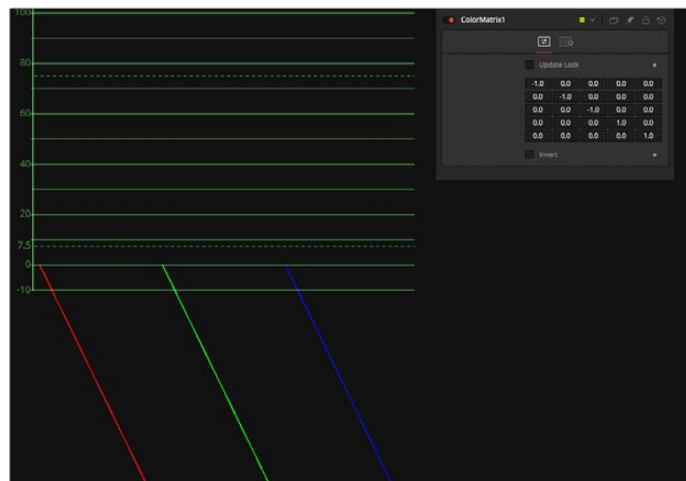
Let's follow this example step by step by viewing the waveform of a 32-bit grayscale gradient.

- 1 The original grayscale.



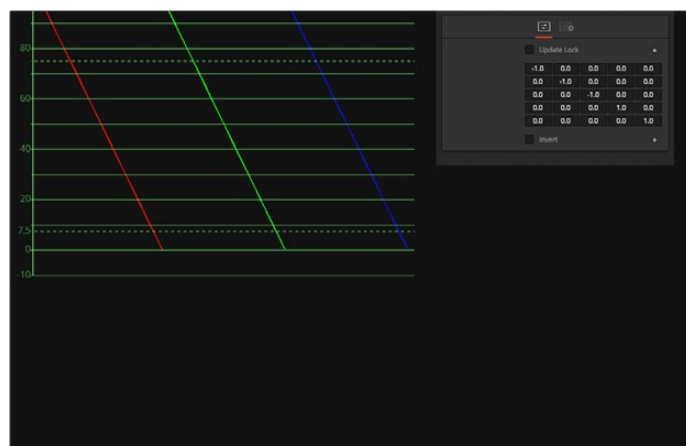
Original Grayscale

- 2 RGB set to -1. The values get inverted but fall below 0.



RGB set to -1

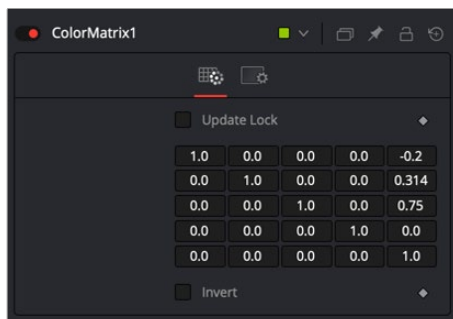
- 3 Adding 1 to each channel keeps the inversion but moves the values back into a positive range.



Adding 1 to each channel

Example 2: Brightness per Channel

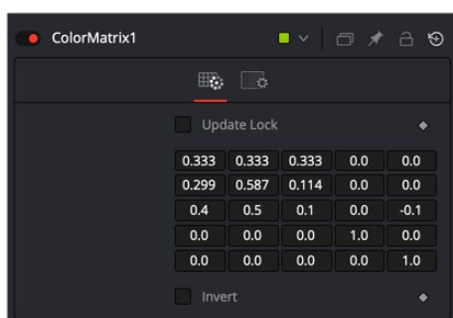
This example influences the brightness of each channel individually. This subtracts 0.2 from the red channel, adds 0.314 to the green channel, and adds 0.75 to the blue channel, while keeping Alpha as it is.



Brightness per channel example

Example 3: Copying Values

You can also copy color values back and forth between individual channels. In this example, the red channel contains the luminance values of the image based on thirds, and the green channel contains the luminance values based on the proper black-and-white conversion method, whereas the blue channel uses a third method based on getting more information from red and less from blue. The blue channel's brightness is also lowered by 0.1, and the Alpha channel is replaced with the original blue channel.

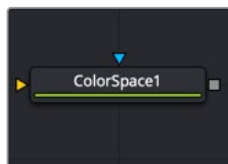


Copying Values example

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Color Space [CS]



The Color Space node

Color Space Node Introduction

The Color Space node provides the ability to work on an image in a variety of alternate color space formats. By default, Fusion uses the RGB color space, and most nodes and displays interpret the primary channels of an image as Red, Green, and Blue.

Changing the color space from RGB causes most images to look odd, as Fusion's viewers still interpret the primary channels as Red, Green, and Blue. For example, viewing an image converted to YUV in one of the viewers shows the Y channel as Red, the U channel as Green, and the V channel as Blue.

Several common elements of the Fusion interface refer to the RGB channels directly. The four buttons commonly found on the Inspector's Settings tab to restrict the effect of the node to a single color channel are one example. When a conversion is applied to an image, the labels of these buttons remain R, G, and B, but the values they represent are from the current color space. (For example, Red is Hue, Green is Luminance, and Blue is Saturation for an RGB to HLS conversion. The Alpha value is never changed by the color space conversion.)

Inputs

The Color Space node includes two inputs: one for the main image and the other for an effect mask.

Input: This orange input is the only required connection. It connects a 2D image that is converted by the color space operation.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color space adjustment to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

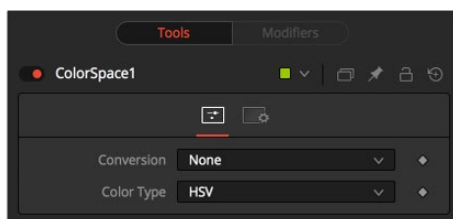
Basic Node Setup

The Color Space node, like many 2D image-processing nodes, receives a 2D image like a Loader or the Medialn1 shown below, processes the image, and then extends the node tree by connecting to another 2D image-processing node or a Merge node.



A Color Space node applies a conversion to a Medialn1 node

Inspector



Color Space controls

Controls Tab

The Controls tab in the Color Space node consists of two menus. The top Conversion menu determines whether you are converting an image to RGB or from RGB. The bottom menu selects the alternative color space you are either converting to or from.

Conversion

This menu has three options. The None option has no effect on the image. When To Color is selected, the input image is converted to the color space selected in the Color Type control found below. When To RGB is selected, the input image is converted back to the RGB color space from the type selected in the Color Type menu (for example, YUV to RGB).

Color Type

This menu is used to select the color space conversion applied when the To Color conversion is selected. When the To RGB option is selected in the Conversion menu, the Color Type option should reflect the input image's current color space. There are eight color space options to choose from.

- **HSV (Hue, Saturation, and Value):** Each pixel in the HSV color space is described in terms of its Hue, Saturation, and Value components. Value is defined as the quality by which we distinguish a light color from a dark one or brightness. Decreasing saturation roughly corresponds to adding white to a paint chip on a palette. Increasing Value is roughly similar to adding black.
- **YUV (Luma, Blue Chroma, and Red Chroma):** The YUV color space is used in the analog broadcast of PAL video. Historically, this format was often used to color correct images because of its familiarity to a large percentage of video engineers. Each pixel is described in terms of its Luminance, Blue Chroma, and Red Chroma components.
- **YIQ (Luma, In Phase, and Quadrature):** The YIQ color space is used in the analog broadcast of NTSC video. This format is much rarer than YUV and almost never seen in production. Each pixel is described in terms of its Luminance, Chroma (in-phase or red-cyan channel), and Quadrature (magenta-green) components.
- **CMY (Cyan, Magenta, and Yellow):** Although more common in print, the CMY format is often found in computer graphics from other software packages. Each pixel is described in terms of its Cyan, Magenta, and Yellow components. CMY is nonlinear.
- **HLS (Hue, Luminance, and Saturation):** Each pixel in the HLS color space is described in terms of its Hue, Luminance, and Saturation components. The differences between HLS and HSV color spaces are minor.
- **XYZ (CIE Format):** This mode is used to convert a CIE XYZ image to and from RGB color spaces. CIE XYZ is a weighted space, instead of a nonlinear one, unlike the other available color spaces. Nonlinear in this context means that equal changes in value at different positions in the color space may not necessarily produce the same magnitude of change visually to the eye.

Expressed simply, the CIE color space is a perceptual color system, with weighted values obtained from experiments where subjects were asked to match an existing light source using three primary light sources.

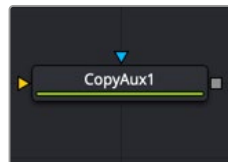
This color space is most often used to perform gamut conversion and color space matching between image display formats because it contains the entire gamut of perceivable colors.

- **Negative:** The color channels are inverted. The color space remains RGBA.
- **BW:** The image is converted to black and white. The contribution of each channel to the luminance of the image is adjustable via slider controls that appear when this option is selected. The default values of these sliders represent the usual perceptual contribution of each channel to an image's luminance. The color space of the image remains RGBA.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Copy Aux [CpA]



The Copy Aux node

Copy Aux Node Introduction

The Copy Aux node is used to shuffle channels between visible channels and auxiliary data channels in a single 2D image. Typically, these auxiliary channels are rendered from 3D applications. Auxiliary channels supported in the Copy Aux node include background color, z-depth, texture coordinates, coverage, object ID, material ID, normals, vectors, back vectors, and world position.

The Copy Aux node is mostly a convenience node, as the copying can also be accomplished with more effort using a Channel Booleans node. Where Channel Booleans deals with individual channels, Copy Aux deals with channel groups. By default, the Copy Aux node automatically promotes the depth of its output to match the depth of the aux channel.

Copy Aux also supports static normalization ranges. The advantage of static normalization versus the dynamic normalization that Fusion's viewers do is that colors remain constant over time. For example, if you are viewing Z or WorldPos values for a ball, you see a smooth gradient from white to black. Now imagine that some other 3D object is introduced into the background at a certain time. Dynamic normalization turns the ball almost completely white while the background object is now the new black. Dynamic normalization also causes flicker problems while viewing vector/disparity channels, which can make it difficult to compare the aux channels of two frames at different times visually.

Inputs

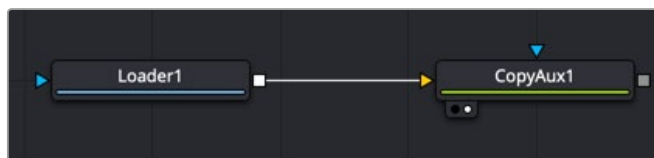
The Copy Aux node includes two inputs: one for the main image and the other for an effect mask.

Input: This orange input is the only required connection. It connects a 2D image for the Copy Aux node operation.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Copy Aux operation to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

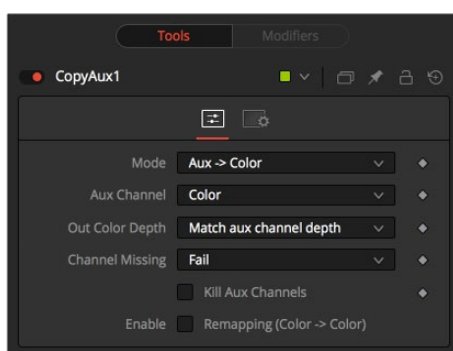
Basic Node Setup

The Copy Aux node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Copy Aux node applied to a MediaIn1 node

Inspector



Copy Aux controls

Controls Tab

The Controls tab is used to copy auxiliary channel groups into RGBA channels. Although Copy Aux has quite a few options, most of the time you select only the channel to copy and ignore the remaining functionality.

Mode

The Mode menu determines whether the auxiliary channel is copied into the RGBA color channel (Aux to Color) or vice versa (Color to Aux). Using this option, you can use one Copy Aux node to bring an auxiliary channel into color, do some compositing operations on it, and then use another Copy Aux node to write the color back into the auxiliary channel. When the Mode is set to Color to Aux, all the options in the Controls tab except the Aux Channel menu are hidden.

Aux Channel

The Aux Channel menu selects the auxiliary channel to be copied from or written to depending on the current mode. When the aux channel abcd has one valid component, it is copied as aaa1, two valid components as ab01, three valid components as abc1, and four components as abcd. For example, the Z-channel is copied as zzz1, texture coordinates as uv01, and normals as nxnynz1.

Out Color Depth

Out Color Depth controls the color depth of the output image. Most aux channels contain float values or, if they are integer valued, they can contain values beyond 255. When you copy float values into an int8 or int16 image, this can be a problem since negative values and values over 1.0 can get clipped. In addition, precision can be lost. This option determines what happens if the depth of RGBA channels of the input image is insufficient to contain the copied aux channel.

Be careful when copying float channels into integer image formats, as they can get clipped if you do not set up Copy Aux correctly. For this node, all aux channels are considered to be float32 except ObjectID or MaterialID, which are considered to be int16.

- **Match Aux Channel Depth:** The bit depth of the RGBA channels of the output image is increased to match the depth of the aux channel. Specifically, this means that the RGBA channels of the output image are either int16 or float32. Be careful when using this option because, for example, if you normally have int8 color channels, you are now using 2x or 4x more memory for the color channels. Particularly, the Z, Coverage, TextureCoordinate, Normal, Vector, BackVector, WorldPosition, and Disparity channels are always output as float, and the Material/ObjectID channels are output as int16.
- **Match Source Color Depth:** The bit depth of the RGBA channels of the output image is the same as the input image. This can have some unexpected consequences. For example, if your input image is int8, the XYZ components of normals that are floating-point numbers in the [-1, 1] range are clipped to non-negative numbers [0, 1] range. As a more extreme example, consider what happens to Z values. Z values are floating-point numbers stored in the [-1e30, 0] range, and they all get truncated to the [0, 1] range, which means your Z-channel is full of zeroes.
- **Force Float32:** The bit depth of the RGBA channels of the output image is always float32.

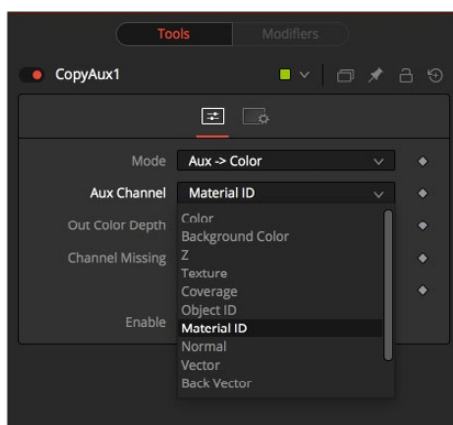
Channel Missing

Channel Missing determines what happens if a channel is not present. For example, this determines what happens if you chose to copy Disparity to Color and your input image does not have a Disparity aux channel.

- **Fail:** The node fails and prints an error message to the console.
- **Use Default Value:** This fills the RGBA channels with the default value of zero for everything except Z, which is -1e30.

Kill Aux Channels

When this is checked, Copy Aux copies the requested channel to RGBA and then outputs a resulting image that is purely RGBA with other channels being killed. This is useful if you want to increase the number of frames of Copy Aux that can be cached for playback—for example, to play back a long sequence of disparity. A handy tip is that you can use the Kill Aux feature also with just Color to Aux > Color for a longer color playback.



Kill Aux channels

Enable Remapping

When remapping is enabled, the currently selected aux channel is rescaled, linearly mapping the range according to the From and To slider selections as explained below. The Remapping options are applied before the conversion operation. This means you could set the From > Min-Max values to -1, 1 to rescale your normals into the [0, 1] range, or set them to [-1000, 0] to rescale your Z values from [-1000, 0] into the [0, 1] range before the clipping occurs.

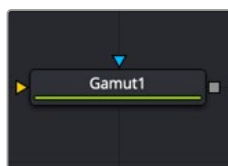
Note that the Remapping options are per channel options. That means the default scale for normals can be set to [-1, +1] > [0, 1] and for Z it can be set [-1000, 0] > [0, 1]. When you flip between normals and Z, both options are remembered. One way this could be useful is that you can set up the remapping ranges and save this as a setting that you can reuse. The remapping can be useful to squash the aux channels into a static [0, 1] range for viewing or, for example, if you wish to compress normals into the [0, 1] range to store them in an int8 image.

- **From > Min:** This is the value of the aux channel that corresponds to To > Min.
- **From > Max:** This is the value of the aux channel that corresponds to To > Max. It is possible to set the max value less than the min value to achieve a flip/inversion of the values.
- **Detect Range:** This scans the current image to detect the min/max values and then sets the From > Min/ From > Max Value controls to these values.
- **Update Range:** This scans the current image to detect the min/max values and then enlarges the current [From > Min, From > Max] region so that it contains the min/max values from the scan.
- **To > Min:** This is the minimum output value, which defaults to 0.
- **To > Max:** This is the maximum output value, which defaults to 1.
- **Invert:** After the values have been rescaled into the [To > Min, To > Max] range, this inverts/flips the range.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Gamut [Gmt]



The Gamut node

Gamut Node Introduction

The Gamut node has controls to transform one color space to another and remove/add gamma curves. This node, along with the Cineon Log node, is primarily used to linearize incoming images and then reapply the applicable output gamma curve at the end of a node tree.

Inputs

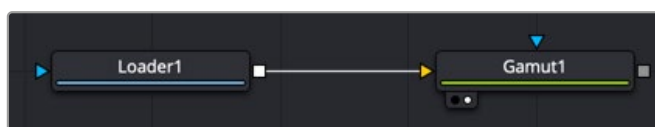
The Gamut node includes two inputs: one for the main image and the other for an effect mask to limit the conversion area.

Input: This orange input is the only required connection. It connects a 2D image output that is the source of the gamut conversion.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Gamut operation to only those pixels within the mask. An effect mask is applied to the tool after the tool is processed.

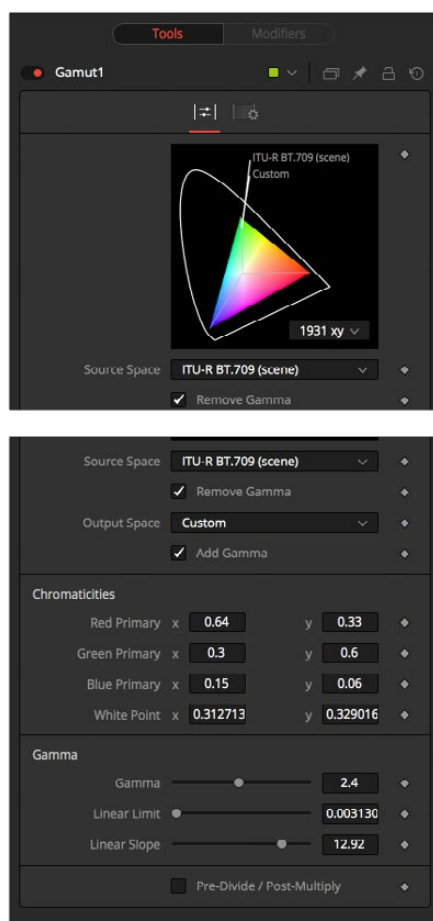
Basic Node Setup

A Gamut node is most often placed directly after the MediaIn node in DaVinci Resolve or a Loader node in Fusion Studio. Another Gamut node is usually placed at the end of a node tree before a MediaOut node in DaVinci Resolve or a Saver node in Fusion Studio.



A Gamut node applied to a MediaIn1 node

Inspector



Gamut controls

Controls Tab

The Controls tab is where all the conversion operations take place. It has a section for incoming images and a section for the node's output. Which section you use depends on whether you are stripping an image of a gamma curve to make it linear or converting a linear image to a specific color space and gamma curve for output.

Source Space

Source Space determines the input color space of the image. When placed directly after a Loader node in Fusion or a MediaIn node in DaVinci Resolve, you would select the applicable color space based on how the image was created and check the Remove Gamma checkbox. The output of the node would be a linearized image. You leave this setting at No Change when you are adding gamma using the Output Space control and placing the node directly before the Saver node in Fusion or a MediaOut node in DaVinci Resolve.

DCI-P3

The DCI-P3 color space is most commonly used in association with DLP projectors. It is frequently provided as a color space available with DLP projectors and as an emulation mode for 10-bit LCD monitors such as the HP Dreamcolor and Apple's Pro Display XDR. This color space is defined in the SMPTE-431-2 standard.

Custom

The Custom gamut allows you to describe the color space according to CIE 1931 primaries and white point, which are expressed as XY coordinates, as well as by gamma, limit, and slope. For example, the DCI-P3 gamut mentioned above would have the following values if described as a Custom color space.

Red Primary	0.68	0.32
Green Primary	0.265	0.69
Blue Primary	0.15	0.06
White Point	0.314	0.351
Gamma	2.6	—
Linear Limit	0.0313	—

To understand how these controls work, you could view the node attached to a gradient background in Waveform mode and observe how different adjustments modify the output.

Output Space

Output Space converts the gamut to the desired color space. For instance, when working with linearized images in a composite, you place the Gamut node just before the Saver node and use the Output Space to convert to the gamut of your final output file. You leave this setting at No Change when you want to remove gamma using the Source Space control.

NOTE: When outputting to HD specification Rec. 709, Fusion uses the term Scene to refer to a gamma of 2.4 and the term Display for a gamma of 2.2.

Remove/Add Gamma

Select these checkboxes to do the gamut conversion in a linear or nonlinear gamma, or simply remove or add the applicable gamma values without changing the color space.

Pre-Divide/Post-Multiply

Selecting this checkbox causes the image's pixel values to be divided by the Alpha values prior to the color correction, and then re-multiplied by the Alpha value after the correction. This helps to avoid the creation of illegally additive images, particularly around the edges of a blue/green key or when working with 3D-rendered objects.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Hue Curves [HCv]



The Hue Curves node

Hue Curves Node Introduction

The Hue Curves node allows you to adjust the color in an image using a series of spline curves. Splines are provided to control the image's hue, saturation, and luminance as well as each individual color channel. An additional set of curves allows you to apply suppression to individual color channels.

The advantage of the Hue Curves node over other color correction nodes in Fusion is that the splines can be manipulated to restrict the node's effect to a very narrow portion of the image, or expanded to include a wide-ranging portion of the image. Additionally, these curves can be animated to follow changes in the image over time. Since the primary axis of the spline is defined by the image's hue, it is much easier to isolate a specific color from the image for adjustment.

Inputs

The Hue Curves node includes two inputs: one for the main image and the other for an effect mask to limit the color correction area.

Input: This orange input is the only required connection. It connects a 2D image for the Hue Curves color correction.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Hue Curves operation to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

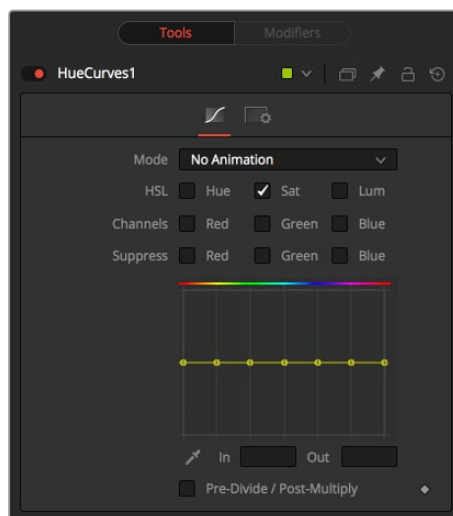
Basic Node Setup

The Hue Curves node, like many 2D image-processing nodes, receives a 2D image like a Loader node or the MediaIn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A Hue Curves node applied to a Loader node in Fusion Studio

Inspector



Hue Curves controls

Controls Tab

The Controls tab consists of color attribute checkboxes that determine which splines are displayed in the Spline window. The spline graph runs horizontally across with control points placed horizontally at each of the primary colors. You can manipulate these control points to change the selected color attribute.

Mode

The Mode options change between No Animation and Animated Points modes. The default mode is No Animation, where adjustments to the curves are applied consistently over time. Setting the Mode to Animated Points or Dissolve allows the color curve to be animated over time.

Dissolve mode is essentially obsolete and is included for compatibility reasons only.

Color Channel Checkboxes

These checkboxes define which splines are editable and are included when using the Eyedropper to pick a color in the image.

Any number of activated splines can be edited simultaneously; however it's more convenient to have only the currently modified spline active to avoid unwanted changes to other splines.

When using the Eyedropper icon, a point is created on all active splines, representing the selected color.

Spline Window

This graph display is the main interface element of the Hue Curves node, which hosts the various splines. In appearance, the node is very similar to the Color Curves node, but here the horizontal axis represents the image's hue, while the vertical axis represents the degree of adjustment. The Spline window shows the curves for the individual channels. It is a miniature Spline Editor. In fact, the curves shown in this window can also be found and edited in the Spline Editor.

The spline curves for all components are initially flat, with control points placed horizontally at each of the primary colors. From left to right, these are: Red, Yellow, Green, Cyan, Blue, and Magenta. Because of the cyclical design of the hue gradient, the leftmost control point in each curve is connected to the rightmost control point of the curve.

Right-clicking in the graph displays a contextual menu containing options for resetting the curves, importing external curves, adjusting the smoothness of the selected control points, and more.

In and Out

Use the In and Out controls to manipulate the precise values of a selected point. To change a value, select a point and enter the In/Out values desired.

Eyedropper

Left-clicking and dragging from the Eyedropper icon changes the current mouse cursor to an Eyedropper. While still holding down the mouse button, drag the cursor to a viewer to pick a pixel from a displayed image. This causes control points, which are locked on the horizontal axis, to appear on the currently active curves. The control points represent the position of the selected color on the curve. Use the contextual menu's Lock Selected Points toggle to unlock points and restore the option of horizontal movement.

Points are only added to enabled splines. To add points only on a specific channel, disable the other channels before making the selection.

Pre-Divide/Post-Multiply

Selecting this checkbox causes the image's pixel values to be divided by the Alpha values prior to the color correction, and then re-multiplied by the Alpha value after the correction. This helps when color correcting images that include a premultiplied Alpha channel.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

OCIO CDL Transform [OCD]



The OCIO CDL Transform node

OCIO CDL Transform Node Introduction

Fusion supports the Open Color IO color management workflow by way of three OCIO nodes.

- The OCIO CDL Transform node allows you to create, save, load, and apply a Color Decision List (CDL) grade.
- The OCIO Color Space allows sophisticated color space conversions, based on an OCIO config file.
- The OCIO File Transform allows you to load and apply a variety of Lookup tables (LUTs).

Generally, the OCIO color pipeline is composed from a set of color transformations defined by OCIO-specific config files, commonly named with a “.ocio” extension. These config files allow you to share color settings within or between facilities. The path to the config file to be used is normally specified by a user-created environment variable called “OCIO,” although some tools allow overriding this. If no other *.ocio config files are located, the DefaultConfig.ocio file in Fusion’s LUTs directory is used.

For in-depth documentation of the format’s internals, please refer to the official pages on openclorio.org.

Inputs

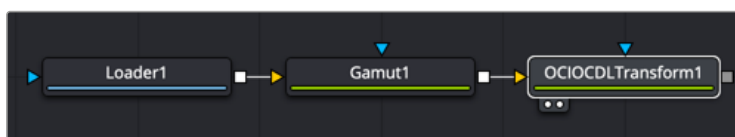
The OCIO CDL Transform node includes two inputs: one for the main image and the other for an effect mask to limit the area where the CDL is applied.

Input: This orange input is the only required connection. It connects a 2D image output for the CDL grade.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the CDL grade to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

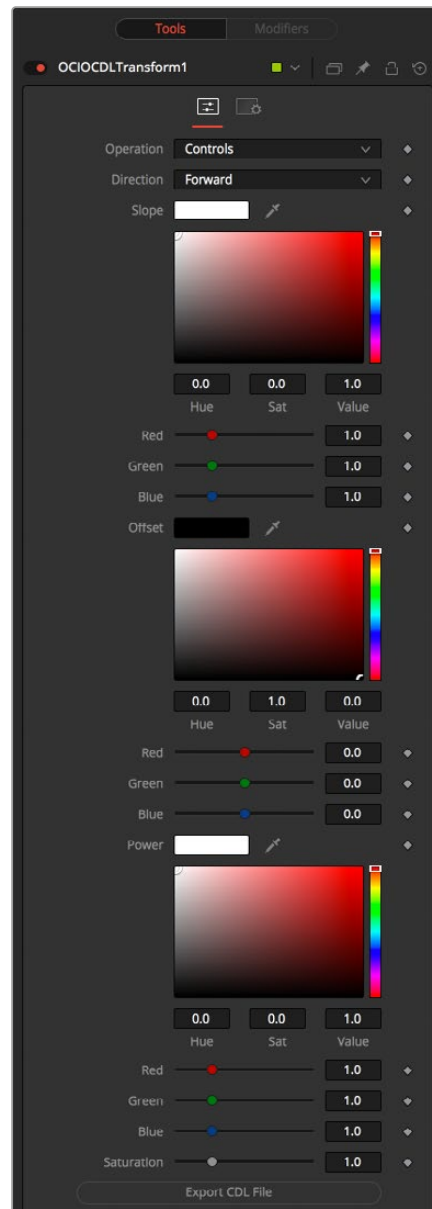
Basic Node Setup

The OCIO CDL Transform node is often applied after a Gamut node converts the Loader to linear color in Fusion Studio.



A OCIO CDL Transform node applied to a Loader node after a Gamut conversion to linear in Fusion Studio

Inspector



OCIO Transform controls

Controls Tab

The Controls tab for the OCIO CDL Transform contains primary color grading color correction controls in a format compatible with CDLs. You can make R, G, B adjustments based on the Slope, Offset, and Power. There is also overall Saturation control. You can also use the Controls tab to import and export the CDL compatible adjustments.

Operation

This menu switches between File and Controls. In File mode, standard ASC-CDL files can be loaded. In Controls mode, manual adjustments can be made to Slope, Offset, Power, and Saturation, and the CDL file can be saved.

NOTE: Using DaVinci Resolve terminology, slope is similar to gain. It controls mids-to-high contrast. Offset is the overall offset of color balance and exposure. Power is very similar to contrast with a raised pivot, giving you control over shadow contrast.

Direction

Toggles between Forward and Reverse. Forward applies the corrections specified in the node, while Reverse tries to remove those corrections. Keep in mind that not every color correction can be undone.

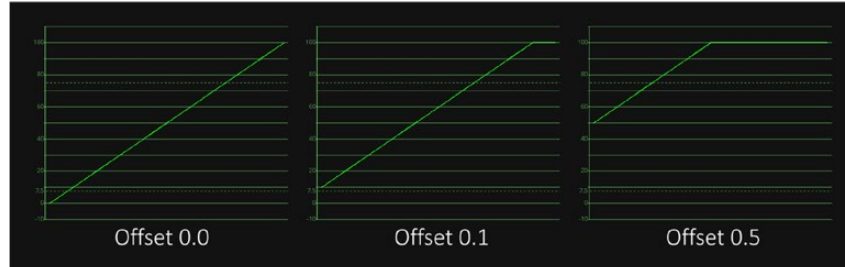
Imagine that all slope values have been set to 0.0, resulting in a fully black image. Reversing that operation is not possible, neither mathematically nor visually.

Slope



Slope multiplies the color values; this is the same as Gain in the Brightness Contrast node.

Offset



Offset adds to the color values; this is the same as Brightness in the Brightness Contrast node.

Power

Applies a Gamma Curve. This is an inverse of the Gamma function of the Brightness Contrast node.

Saturation

Enhances or decreases the color saturation. This works the same as Saturation in the Brightness Contrast node.

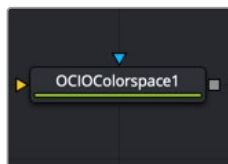
Export File

Allows the user to export the settings as a CDL file.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

OCIO Color Space [OCC]



The OCIO Color Space node

OCIO Color Space Node Introduction

Fusion supports the Open Color IO color management workflow by way of three OCIO nodes.

- The OCIO CDL Transform node allows you to create, save, load, and apply a Color Decision List (CDL) grade.
- The OCIO Color Space allows sophisticated color space conversions, based on an OCIO config file.
- The OCIO File Transform allows you to load and apply a variety of Lookup tables (LUTs).

Generally, the OCIO color pipeline is composed from a set of color transformations defined by OCIO-specific config files, commonly named with a “.ocio” extension. These config files allow you to share color settings within or between facilities. The path to the config file to be used is normally specified by a user-created environment variable called “OCIO,” though some tools allow overriding this. If no other *.ocio config files are located, the DefaultConfig.ocio file in Fusion’s LUTs directory is used.

For in-depth documentation of the format’s internals, please refer to the official pages on openclorio.org.

Sample configs can be obtained from https://openclorio.readthedocs.io/en/latest/configurations/_index.html#configurations

The functionality of the OCIO Color Space node is also available as a View LUT node from the View LUT menu.

Inputs

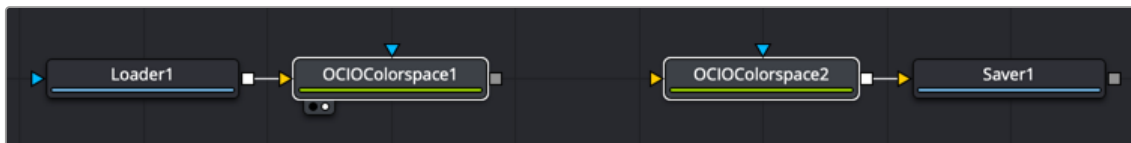
The OCIO Color Space node includes two inputs: one for the main image and the other for an effect mask to limit the area where the color space conversion is applied.

Input: This orange input is the only required connection. It connects a 2D image for the color space conversion.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the color space conversion to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

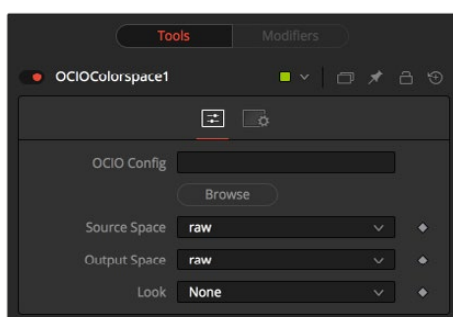
Basic Node Setup

The OCIO Color Space node is typically placed directly after a MediaIn node in DaVinci Resolve or a Loader node in Fusion Studio. Another OCIO Color Space node is placed just before a Media Out node in DaVinci Resolve or a Saver node in Fusion Studio.



An OCIO Color Space node applied to a Loader node and a Saver node in Fusion Studio

Inspector



OCIO Color Space controls

Controls Tab

The Controls tab for the OCIO Color Space node allows you to convert an image from one color space to another based on an OCIO config file. By default, it uses the config file included with Fusion; however, the Controls tab does allow you to load your own config file as well.

OCIO Config

Displays a File > Open dialog to load the desired config file.

Source Space

Based on the config file, the available source color spaces are listed here.

The content of this list is based solely on the loaded profile and hence can vary immensely. If no other OCIO config file is loaded, the DefaultConfig.ocio file in Fusion's LUTs directory is used to populate this menu.

Output Space

Based on the config file, the available output color spaces are listed here.

The content of this list is based solely on the loaded profile and hence can vary immensely. If no other OCIO config file is loaded, the DefaultConfig.ocio file in Fusion's LUTs directory is used to populate this menu.

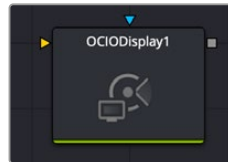
Look

Installed OCIO Color Transform Looks appear in this menu. If no looks are installed, this menu has only None listed as an option.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

OCIO Display [OCD]



The OCIO Display node

OCD Display Node Introduction

The OCIO Display node allows you to load an OpenColorIO (OCIO) Display transform (.ocio) to an input.

Inputs

The single input on the OCD Display node is used to connect the node you want to apply the OCIO Display transform to, and an effect mask, which can be used to limit the display transform area.

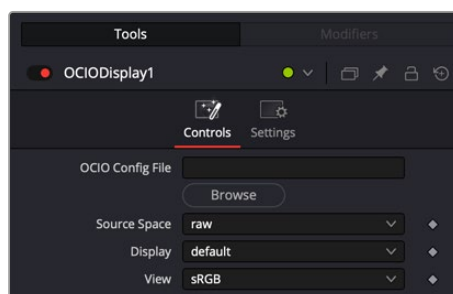
Basic Node Setup

Connect the media or tool that you wish to apply the OCIO Display transform to the input. The output of the node then gets passed to the remainder of the node tree.



A simple node tree showing the OCIO Display node taking the Media In Node, performing the Display Transform, and passing it to Media Out

Inspector



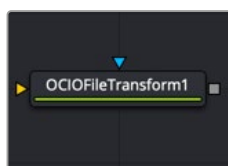
The OCIO Display controls

Controls

The OCIO Display controls lets you load and apply an .ocio configuration file and apply its color space and display transform to to the node's input.

- **OCIO Config File:** Click the Browse button to open up a file system prompt to locate the .ocio file.
- **Source Space:** Choose the color space of the source material from the list.
- **Display:** Choose which monitor your viewer is on.
- **View:** Choose which color space to view.

OCIO File Transform [OCF]



The OCIO File Transform node

OCIO File Transform Node Introduction

Fusion supports the Open Color IO color management workflow by way of three OCIO nodes.

- The OCIO CDL Transform node allows you to create, save, load, and apply a Color Decision List (CDL) grade.
- The OCIO Color Space allows sophisticated color space conversions, based on an OCIO config file.
- The OCIO File Transform allows you to load and apply a variety of Lookup tables (LUTs).

Generally, the OCIO color pipeline is composed from a set of color transformations defined by OCIO-specific config files, commonly named with a “.ocio” extension. These config files allow you to share color settings within or between facilities. The path to the config file to be used is normally specified by a user-created environment variable called “OCIO,” though some tools allow overriding this. If no other *.ocio config files are located, the DefaultConfig.ocio file in Fusion's LUTs directory is used.

For in-depth documentation of the format's internals, please refer to the official pages on openclorio.org.

The functionality of the OCIO File Transform node is also available as a View LUT node from the View LUT menu.

Inputs

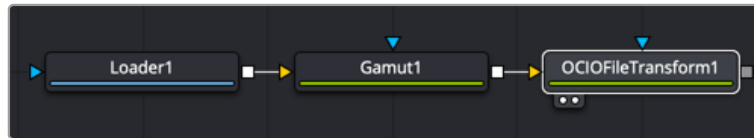
The OCIO File Transform node includes two inputs: one for the main image and the other for an effect mask to limit the area where the color space conversion is applied.

Input: This orange input is the only required connection. It connects a 2D image for the LUT.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the applied LUT to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

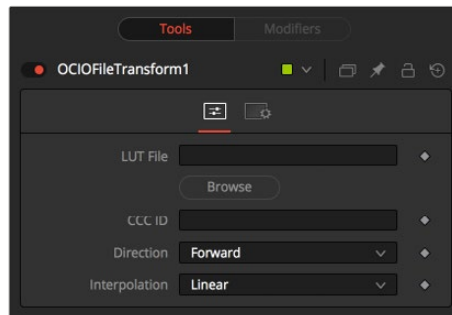
Basic Node Setup

The OCIO File Transform node is often applied after a Gamut node converts the Loader to linear color in Fusion Studio.



An OCIO File Transform node applied to a Loader node after a Gamut node conversion to linear color space

Inspector



OCIO File Transform controls

Controls Tab

The Controls tab for the OCIO File Transform node includes options to import the LUT, invert the transform, and select the color interpolation method.

LUT File

Displays a File > Open dialog to load the desired LUT.

CCC ID

This is the ID key used to identify the specific file transform located within the ASC CDL color correction XML file.

Direction

Toggles between Forward and Reverse. Forward applies the corrections specified in the node, while Reverse tries to remove those corrections. Keep in mind that not every color correction can be undone. Imagine that all slope values have been set to 0.0, resulting in a fully black image. Reversing that operation is not possible, neither mathematically nor visually.

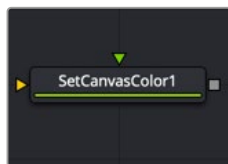
Interpolation

Allows the user to select the color interpolation to achieve the best quality/render time ratio. Nearest is the fastest interpolation, while Best is the slowest.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Set Canvas Color [SCv]



The Set Canvas Color node

Set Canvas Color Node Introduction

Set Canvas Color is used to set the color of the area outside the domain of definition (DoD). This is the workspace area beyond the raster by default, which is invisible since outside the raster is not rendered. However, the DoD can be within the raster as well. This can occur when compositing images smaller than the raster, or with transforms. By default, the canvas color used is black/no Alpha (transparent). However, since some nodes may change an image's canvas color—for example, inverting a mask changes the mask's canvas from black to white—the Set Canvas Color allows you to control the color of the canvas to whatever you require.

The Set Canvas Color node sets the color of the workspace outside the domain of definition (DOD). For example, if you create a circular gradient, the DoD is a square around the circular gradient in the viewer. Everything outside the DoD is understood to be black and therefore does not have to be rendered. To change the area outside the DoD, attach the Set Canvas Color node after the background and change the color.

NOTE: Position the mouse pointer in a black area outside the raster to view the RGB canvas color in the status bar at the bottom left of the Fusion window.

Inputs

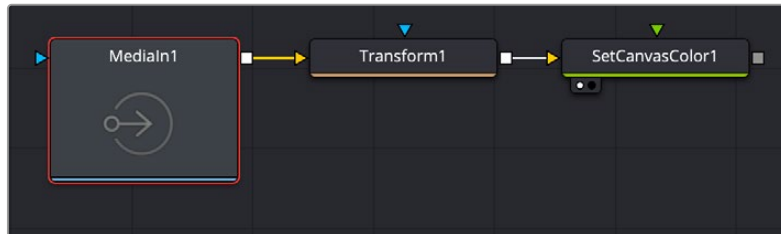
The Set Canvas Color node includes two inputs: one for the main image and a second for a foreground.

Input: This orange input is the only required connection. It accepts a 2D image that reveals the canvas color if the image's DoD is smaller than the raster.

Foreground: The optional green foreground input allows the canvas color to be sampled from an image connected to this input.

Basic Node Setup

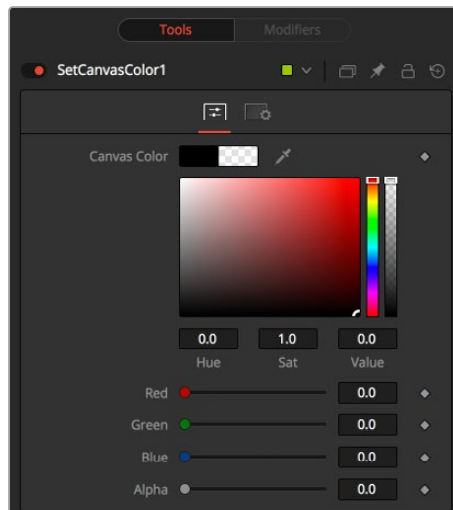
The Set Canvas Color node is placed after the image is transformed to reveal part of the raster outside the domain of definition.



A Set Canvas Color node applied to a transformed MediaIn 1 node

In this example, the footage from MediaIn1 has been shrunk in size using the Transform1 node, leaving empty space around the image in the frame. Feeding that output into the SetCanvasColor node allows you to change the color of the empty space around the frame.

Inspector



Set Canvas Color controls

Controls Tab

The Controls tab for the Set Canvas Color is used for simple color selection. When the green foreground is connected, the tab is empty.

Color Picker

Use these controls to adjust the Color and the Alpha value for the image's canvas. It defaults to black with zero Alpha.

Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

White Balance [WB]



The White Balance node

White Balance Node Introduction

The White Balance node can be used to automatically remove color casts in the image caused by the incorrect setup of a camera or bad lighting conditions.

Correction can be done by selecting a color temperature or by choosing a neutral color from the original image that exhibits the color cast to be corrected.

IMPORTANT: When picking neutral colors using the Custom method, make sure you are picking from the source image, not the results of the White Balance node. This ensures that the image doesn't change while you are still picking, and that the White Balance node gets an accurate idea of the original colors it needs to correct.

Inputs

The White Balance node includes two inputs: one for the main image and the other for an effect mask to limit the area where the white balance is applied.

Input: This orange input is the only required connection. It connects a 2D image for the white balance.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the white balance to only those pixels within the mask. An effect mask is applied to the tool after it is processed.

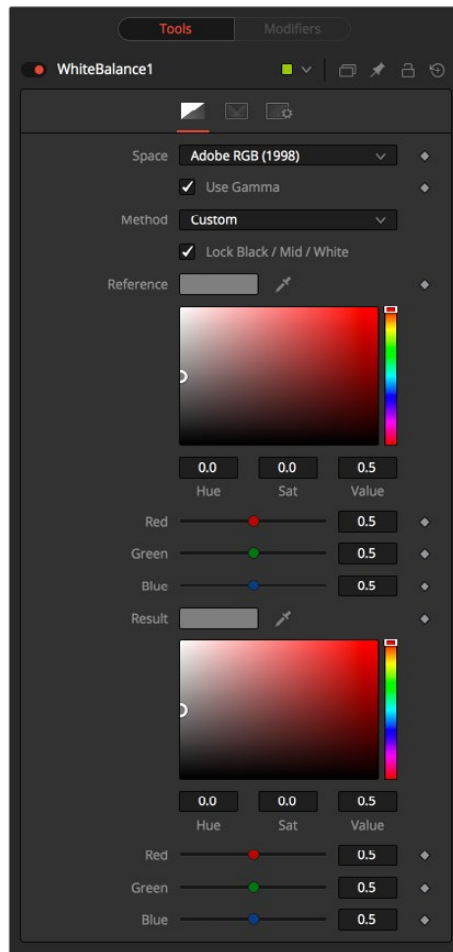
Basic Node Setup

The White Balance node, like many 2D image-processing nodes, receives a 2D image like the Medialn1 shown below. The output continues the node tree by connecting to another 2D image-processing node or a Merge node.



A White Balance node applied to a Medialn1 node

Inspector



White Balance controls

Balance Tab

Space

Use this menu to select the color space of the source image, if it is known. This can make the correction more accurate since the node can take the natural gamma of the color space into account as part of the correction. If the color space that the image uses is unknown, leave this menu at its default value.

Method

The White Balance node can operate using one of two methods: a Custom method or a color Temperature method.

- **Custom:** The Custom method requires the selection of a pixel from the scene that should have been pure gray. The node uses this information to calculate the color correction required to convert the pixel so that it actually is gray. When the correction is applied without an effect mask connected and the LockBlack/Mid/White checkbox enabled, the node white balances the entire shot.
- **Temperature:** The color Temperature method requires that the actual color temperature of the shot be specified.

Lock Black/Mid/White

This checkbox locks the Black, Midtones, and White points together so that the entire image is affected equally. Unchecking the control provides individual controls for white balancing each range separately. This control affects both methods equally.

Black/Mid/White Reference

These controls appear only if the Custom method is selected. They are used to select a color from a pixel in the source image. The White Balance node color corrects the image so that the selected color is transformed to the color set in the Result Color Picker below. Generally, this is gray. A color that is supposed to be pure gray but is not truly gray for one reason or another should be selected.

If the Lock Black/Mid/White checkbox is deselected, different references can be selected for each color range.

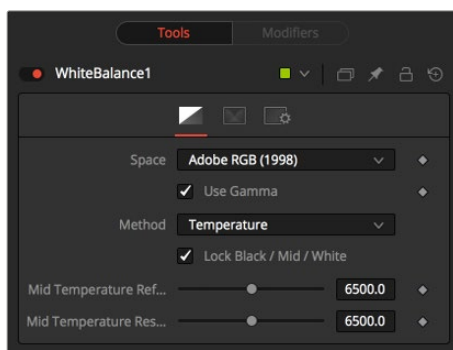
For example, try to select a pixel for the black and white references that are not clipped in any of the color channels. In the high end, an example would be a pixel that is light pink with values of 255, 240, 240. The pixel is saturated/clipped in the red, although the color is not white. Similarly, a really dark blue-gray pixel might be 0, 2, 10. It is clipped in red as well, although it is not black.

Neither example would be a good choice as a reference pixel because there would not be enough headroom left for the White Balance node.

Black/Mid/White Result

These controls appear only if the Custom method is selected. They are used to select the color that the node uses to balance the reference color. This generally defaults to pure, midrange gray.

If the Lock Black/Mid/White checkbox is deselected, different results can be selected for each color range.



White Balance Temperature controls

Temperature Reference

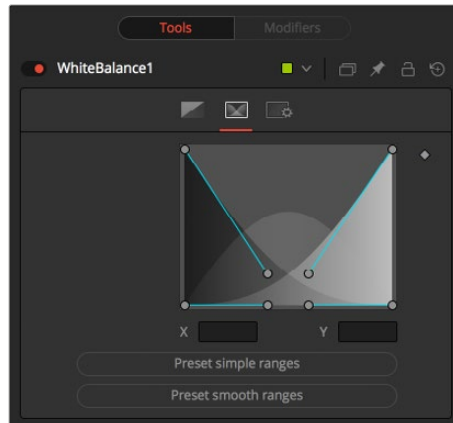
When the Method menu is set to Temperature, the Temperature reference control is used to set the color temperature of the source image. If the Lock Black/ Mid/White checkbox is deselected, different references can be selected for each color range.

Temperature Result

Use this control to set the target color temperature for the image. If the Lock Black/Mid/White checkbox is deselected, different results can be selected for each color range.

Use Gamma

This checkbox selects whether the node takes the gamma of the image into account when applying the correction, using the default gamma of the color space selected in the Space menu at the top of the tab.



White Balance range tab

Ranges Tab

The Ranges tab can be used to customize the range of pixels in the image considered to be shadows, midtones, and highlights by the node.

Spline Display

The ranges are selected by manipulating the spline handles. There are four spline points, each with one Bézier handle. The two handles at the top represent the start of the shadow and highlight ranges, whereas the two at the bottom represent the end of the range. The Bézier handles are used to control the falloff.

The midtones range has no specific controls since its range is understood to be the space between the shadow and the highlight ranges.

The X and Y text controls below the Spline display can be used to enter precise positions for the selected Bézier point or handle.

Preset Simple/Smooth Ranges

These two buttons can be used to return the spline ranges to either Smooth (default) or Simple (linear) settings.

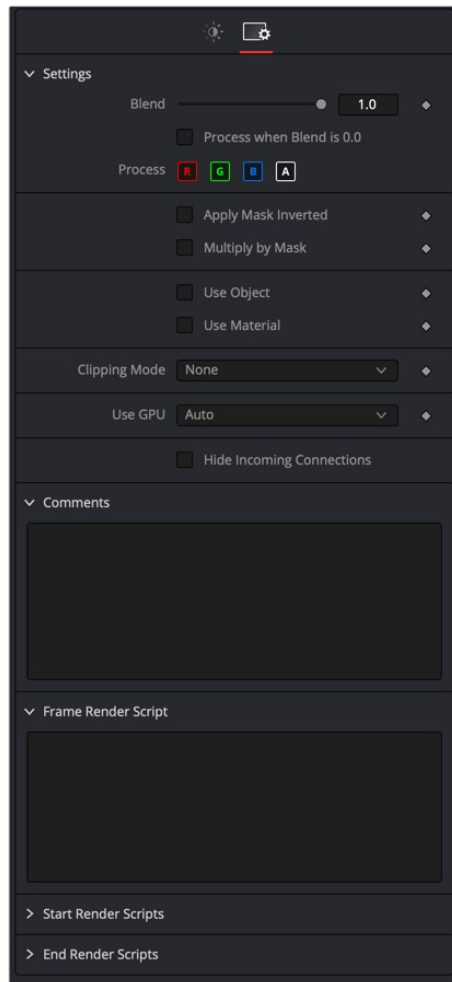
Settings Tab

The Settings tab in the Inspector is also duplicated in other Color nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

The Common Controls

Nodes that handle Color adjustment operations share several identical controls in the Inspector. This section describes controls that are common among color nodes.

Inspector



Color Settings Inspector

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Color category. The Settings controls are even found on third-party color type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options that are also covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is mostly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Comments

The Comments field is used to add notes to a tool. Click in the field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.